



Lecture 7: State Feedback Control System

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SIST 1D#206

Observer Design for Better Disturbance Rejection



- Example paper using frequency responses to evaluate a servo control system

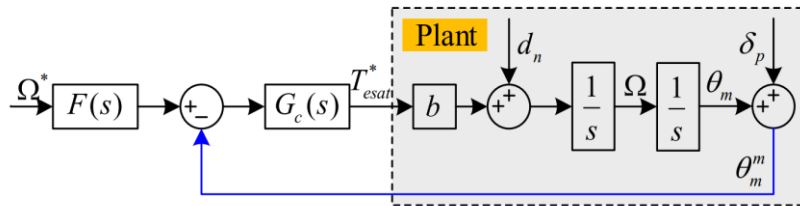


Fig. 3. Block diagram of different ADRC systems in transfer function form

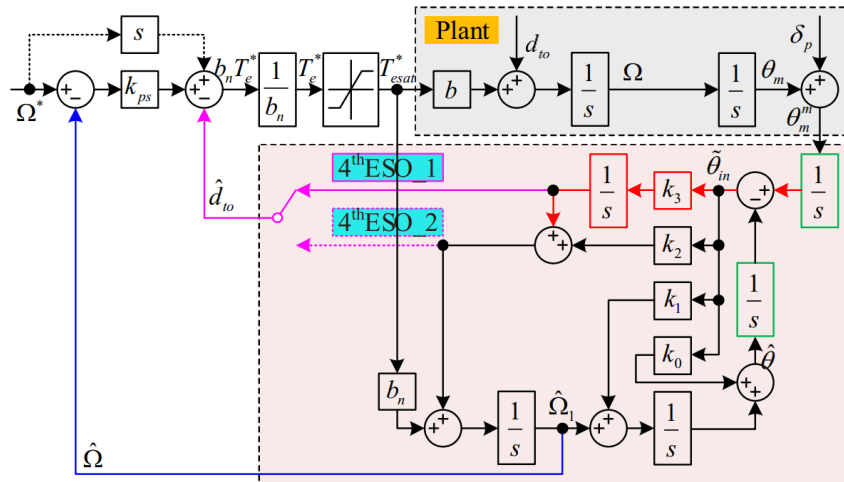


Fig. 14. The equivalent block diagram of the system 4thESO_1-2.

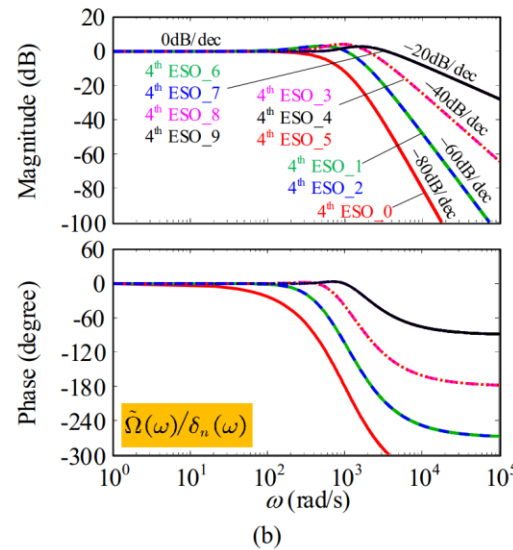
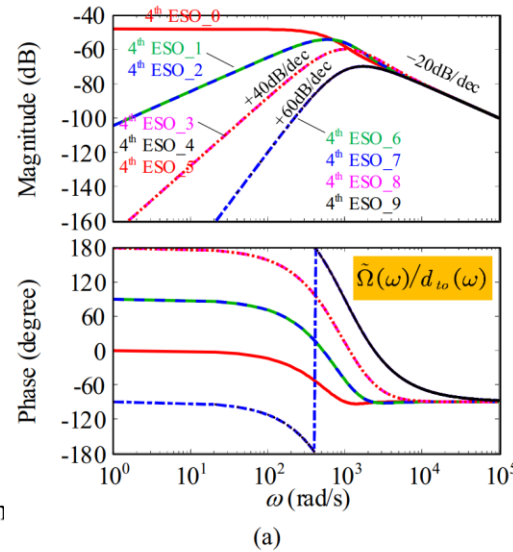


Fig. 10. Bode diagrams of the ten fourth order ESOs for observing speed.

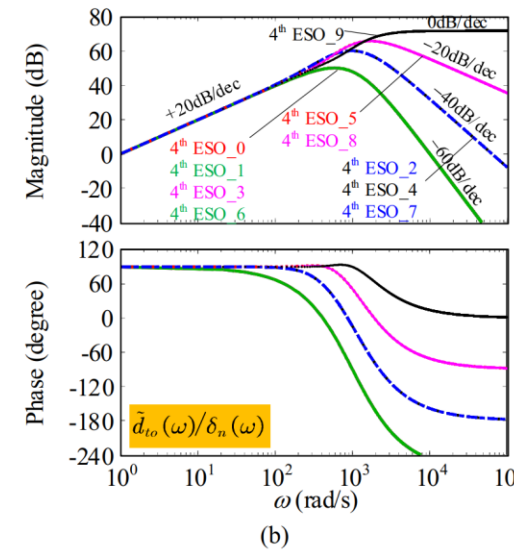
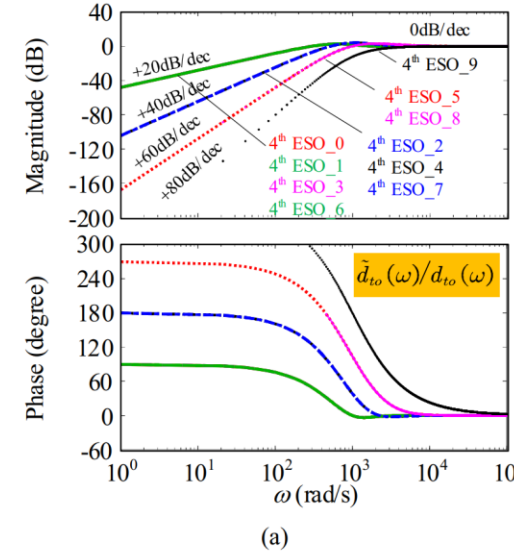


Fig. 11. Bode diagrams of the ten fourth order ESOs for observing disturbance.

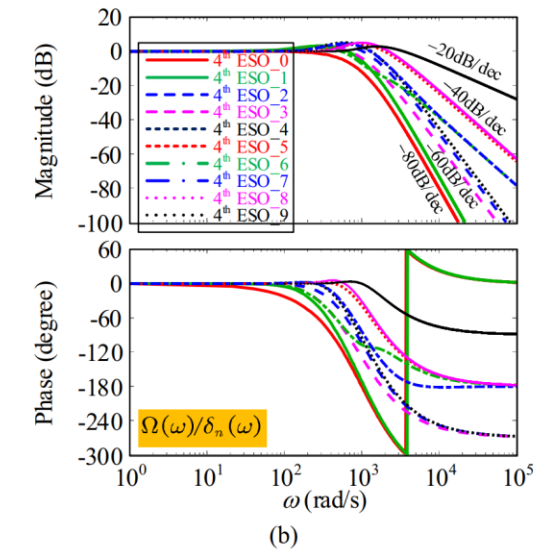
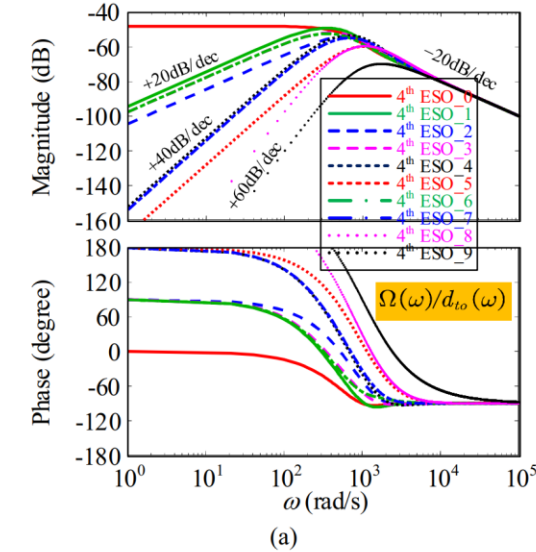


Fig. 12. Bode diagram of the ten systems under the same p_o .



State Space Model (Review)

- The simultaneous **first-order differential** equations about the state variables
 - State variables must be linearly independent; that is, no state variable can be written as a linear combination of the other state variables.

- **Algebraic** output equation

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

$\mathbf{x}$ = state vector

$\dot{\mathbf{x}}$ = derivative of the state vector with respect to time

$\mathbf{y}$ = output vector

$\mathbf{u}$ = input or control vector

$\mathbf{A}$ = system matrix

$\mathbf{B}$ = input matrix

$\mathbf{C}$ = output matrix

$\mathbf{D}$ = feedforward matrix

- E.g.,:

$$\frac{d}{dt} \begin{bmatrix} \Theta \\ \Omega \\ i \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & K_T J^{-1} \\ 0 & -K_E L^{-1} & -R L^{-1} \end{bmatrix} \begin{bmatrix} \Theta \\ \Omega \\ i \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ L^{-1} \end{bmatrix} u + \begin{bmatrix} 0 \\ J^{-1} \\ 0 \end{bmatrix} \tau_L$$

$$y = [1 \ 0 \ 1] \begin{bmatrix} \Theta \\ \Omega \\ i \end{bmatrix}$$

- State space: The n-dimensional space whose axes are the state variables.

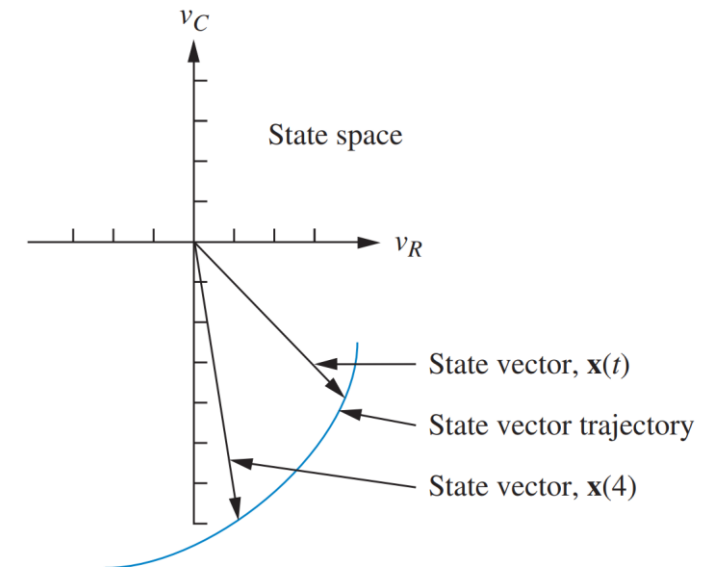


FIGURE 3.3 Graphic representation of state space and a state vector

PROBLEM: Find the state-space representation of the electrical network shown in Figure 3.8. The output is $v_o(t)$.

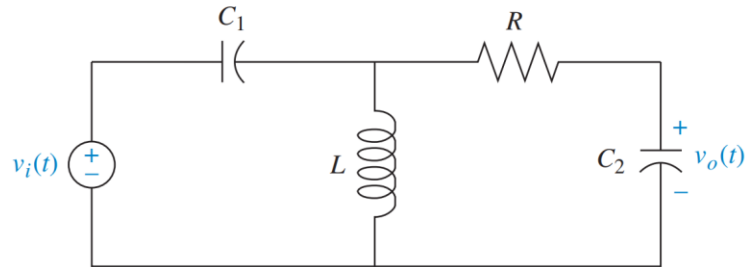


FIGURE 3.8 Electric circuit for Skill-Assessment Exercise 3.1

ANSWER:



PROBLEM: Represent the translational mechanical system shown in Figure 3.9 in state space, where $x_3(t)$ is the output.

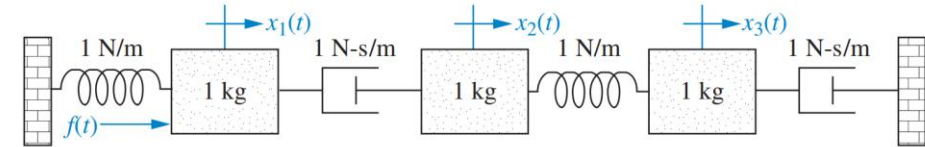


FIGURE 3.9 Translational mechanical system for Skill-Assessment Exercise 3.2

ANSWER:



From O.D.E to State space model



At first we select a set of state variables, called *phase variables*, where each subsequent state variable is defined to be the derivative of the previous state variable.

$$\frac{d^n y}{dt^n} + a_{n-1} \frac{d^{n-1} y}{dt^{n-1}} + \cdots + a_1 \frac{dy}{dt} + a_0 y = b_0 u$$



$$\begin{aligned} x_1 &= y \\ x_2 &= \frac{dy}{dt} \\ x_3 &= \frac{d^2 y}{dt^2} \\ &\vdots \\ x_n &= \frac{d^{n-1} y}{dt^{n-1}} \end{aligned}$$



$$\begin{aligned} \dot{x}_1 &= \frac{dy}{dt} \\ \dot{x}_2 &= \frac{d^2 y}{dt^2} \\ \dot{x}_3 &= \frac{d^3 y}{dt^3} \\ &\vdots \\ \dot{x}_n &= \frac{d^n y}{dt^n} \end{aligned}$$



$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ &\vdots \\ \dot{x}_{n-1} &= x_n \end{aligned}$$



$$\dot{x}_n = -a_0 x_1 - a_1 x_2 \cdots - a_{n-1} x_n + b_0 u$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & \cdots & 0 \\ \vdots & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & -a_3 & -a_4 & -a_5 & \cdots & -a_{n-1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ b_0 \end{bmatrix} u \quad (3.52)$$

$$y = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix}$$

Converting a Transfer Function with a Constant Term in the Numerator

$$\frac{C(s)}{R(s)} = \frac{24}{s^3 + 9s^2 + 26s + 24}$$

$$(s^3 + 9s^2 + 26s + 24)C(s) = 24R(s)$$

$$\ddot{c} + 9\dot{c} + 26c = 24r$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 24 \end{bmatrix} r$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

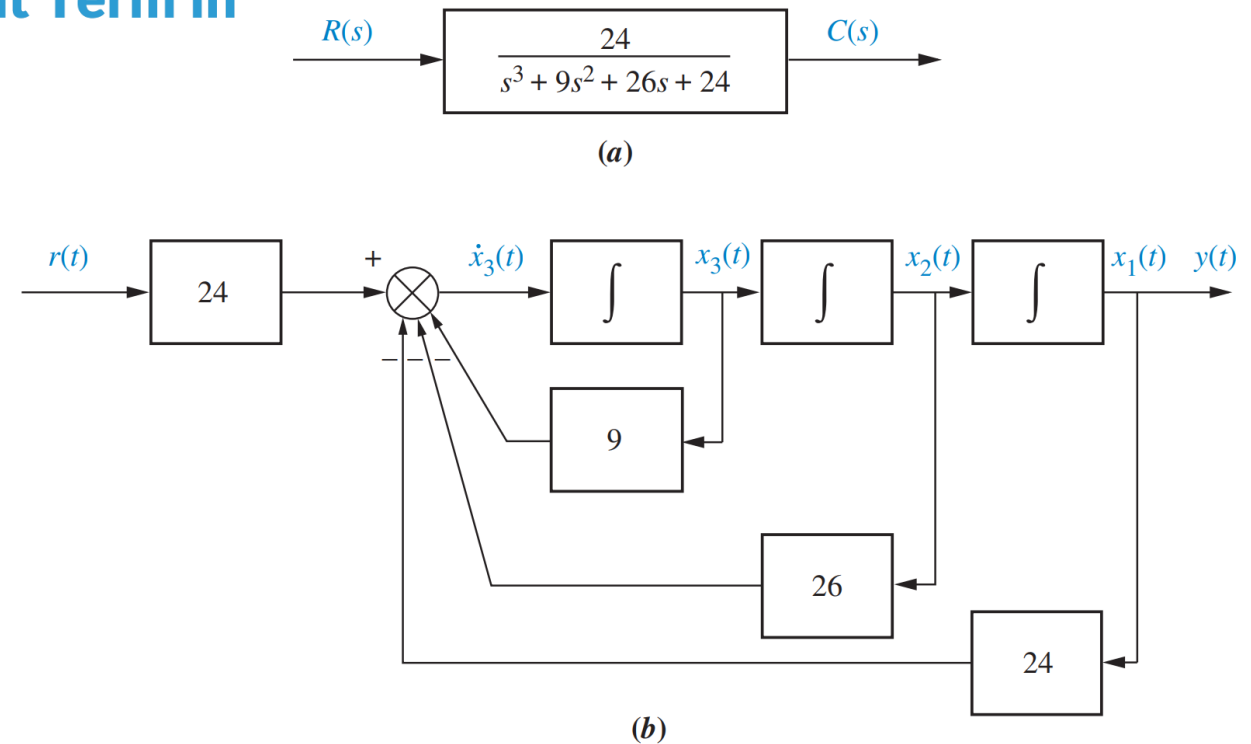


FIGURE 3.10 a. Transfer function; b. equivalent block diagram showing phase variables. Note: $y(t) = c(t)$.

Converting a Transfer Function with a Polynomial in the Numerator

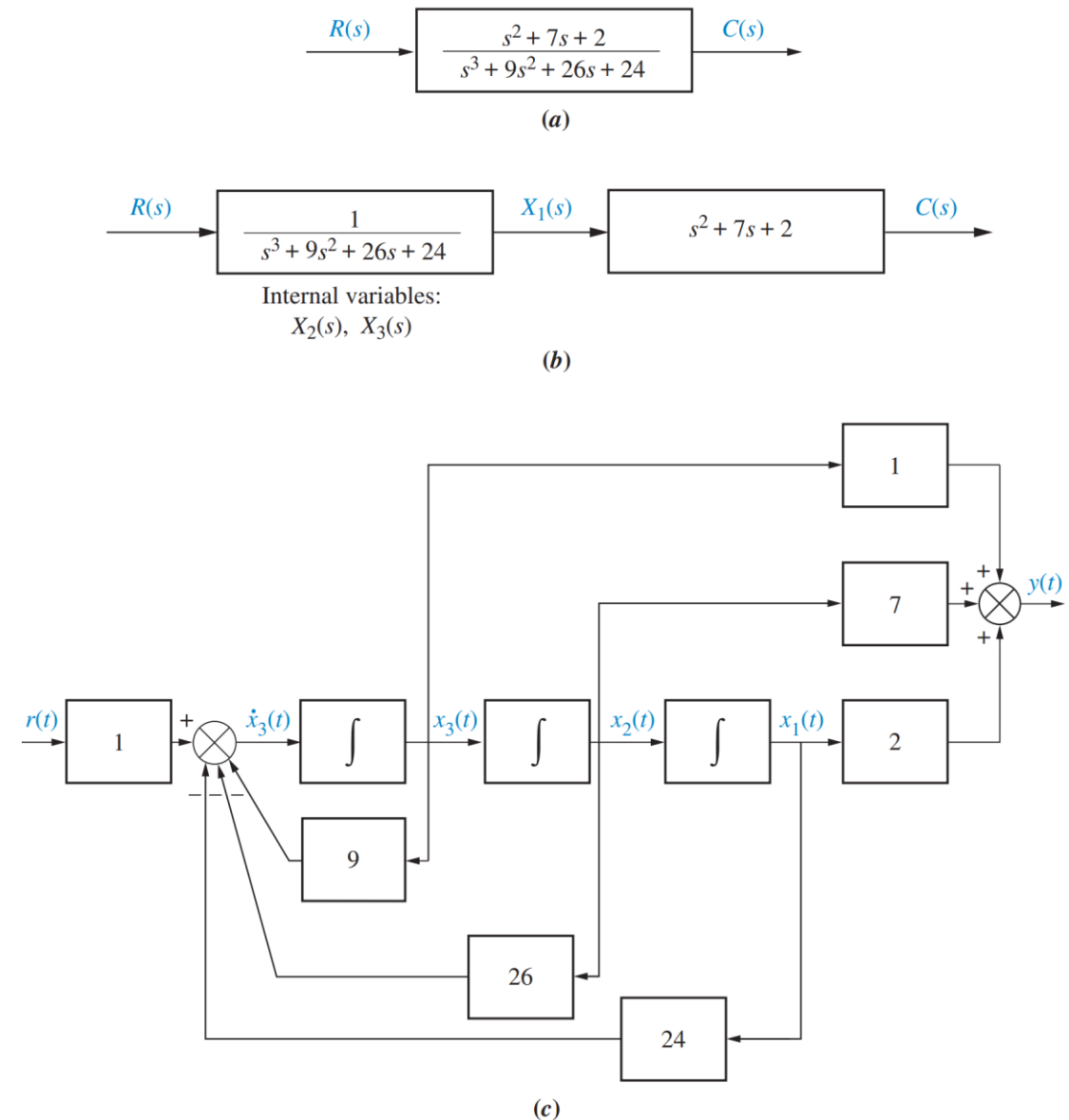


FIGURE 3.12 **a.** Transfer function; **b.** decomposed transfer function; **c.** equivalent block diagram Note: $y(t) = c(t)$.

From transfer function to state space model (2)



PROBLEM: Find the state equations and output equation for the phase-variable representation of the transfer function $G(s) = \frac{2s + 1}{s^2 + 7s + 9}$.

ANSWER:



$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$



$$s\mathbf{X}(s) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}\mathbf{U}(s)$$

$$\mathbf{Y}(s) = \mathbf{C}\mathbf{X}(s) + \mathbf{D}\mathbf{U}(s)$$



$$(s\mathbf{I} - \mathbf{A})\mathbf{X}(s) = \mathbf{B}\mathbf{U}(s)$$



$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{U}(s)$$



$$\mathbf{Y}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{U}(s) + \mathbf{D}\mathbf{U}(s) = [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}]\mathbf{U}(s)$$

$$T(s) = \frac{Y(s)}{U(s)} = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$$



Single input single output

Transfer function matrix

State space model to transfer function (ss2tf)



PROBLEM: Given the system defined by Eq. (3.74), find the transfer function, $T(s) = Y(s)/U(s)$, where $U(s)$ is the input and $Y(s)$ is the output.

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} u \quad (3.74a)$$

$$y = [1 \ 0 \ 0] \mathbf{x} \quad (3.74b)$$

$$T(s) = \frac{Y(s)}{U(s)} = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} + \mathbf{D}$$

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{\text{adj}(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})} = \frac{\begin{bmatrix} (s^2 + 3s + 2) & s + 3 & 1 \\ -1 & s(s + 3) & s \\ -s & -(2s + 1) & s^2 \end{bmatrix}}{s^3 + 3s^2 + 2s + 1} \quad (s\mathbf{I} - \mathbf{A}) = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} = \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 1 & 2 & s + 3 \end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad \text{adj}(\mathbf{A}) = \begin{bmatrix} + \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} & + \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ - \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} & + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} \\ + \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} & + \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \end{bmatrix}^T = \begin{bmatrix} + \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} & + \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} \\ - \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} & + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} \\ + \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} & + \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \end{bmatrix}$$
$$T(s) = \frac{10(s^2 + 3s + 2)}{s^3 + 3s^2 + 2s + 1}$$

TryIt 3.2

Use the following MATLAB and the Control System Toolbox statements to obtain the transfer function shown in Skill-Assessment Exercise 3.4 from the state-space representation of Eq. (3.78).

```
A=[ -4  -1.5; 4  0];  
B=[ 2  0]';  
C=[ 1.5  0.625];  
D=0;  
T=ss(A, B, C, D);  
T=tf(T)
```

PROBLEM: Convert the state and output equations shown in Eq. (3.78) to a transfer function.

$$\dot{\mathbf{x}} = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u(t) \quad (3.78a)$$

$$y = \begin{bmatrix} 1.5 & 0.625 \end{bmatrix} \mathbf{x} \quad (3.78b)$$

ANSWER:



$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{adj}(\mathbf{A}) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

State space model is not unique



$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

$$y = \mathbf{C}\mathbf{x} + \mathbf{D}u$$

$$\mathbf{z} = \mathbf{P}^{-1}\mathbf{x}$$



$$\dot{\mathbf{z}} = \mathbf{P}^{-1}\mathbf{A}\mathbf{P}\mathbf{z} + \mathbf{P}^{-1}\mathbf{B}u$$

$$y = \mathbf{C}\mathbf{P}\mathbf{z} + \mathbf{D}u$$

PROBLEM: Given the system represented in state space by Eqs. (5.73),

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -5 & -7 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \quad (5.73a)$$

$$y = [1 \quad 0 \quad 0] \mathbf{x} \quad (5.73b)$$

transform the system to a new set of state variables, $\mathbf{z}$, where the new state variables are related to the original state variables, $\mathbf{x}$, as follows:

$$\mathbf{z} = \begin{bmatrix} 2 & 0 & 0 \\ 3 & 2 & 0 \\ 1 & 4 & 5 \end{bmatrix} \mathbf{x} = \mathbf{P}^{-1}\mathbf{x} \quad (5.74a)$$

$$z_1 = 2x_1 \quad (5.74b)$$

$$z_2 = 3x_1 + 2x_2 \quad (5.74c)$$

$$z_3 = x_1 + 4x_2 + 5x_3 \quad (5.74c)$$

$$\begin{aligned} \mathbf{P}^{-1}\mathbf{A}\mathbf{P} &= \begin{bmatrix} 2 & 0 & 0 \\ 3 & 2 & 0 \\ 1 & 4 & 5 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -5 & -7 \end{bmatrix} \begin{bmatrix} 0.5 & 0 & 0 \\ -0.75 & 0.5 & 0 \\ 0.5 & -0.4 & 0.2 \end{bmatrix} \\ &= \begin{bmatrix} -1.5 & 1 & 0 \\ -1.25 & 0.7 & 0.4 \\ -2.5 & 0.4 & -6.2 \end{bmatrix} \end{aligned}$$

Diagonal state space representation

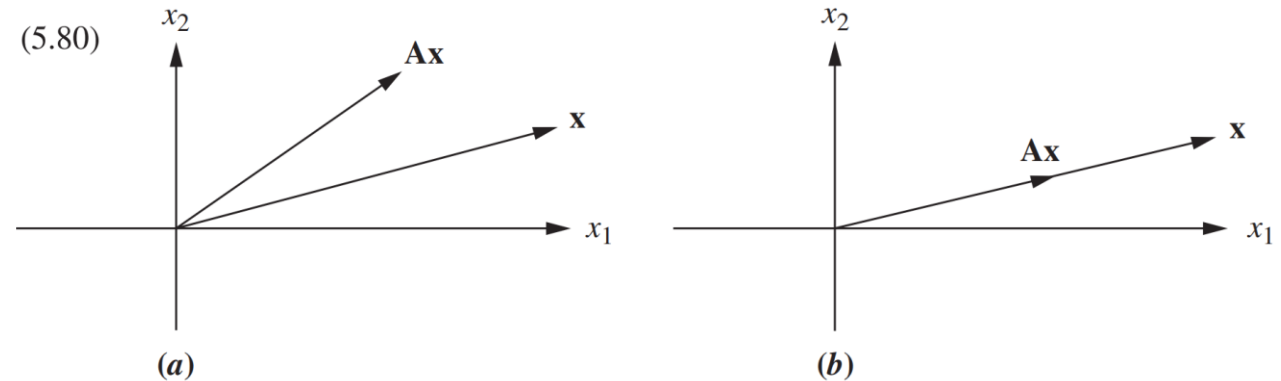


Eigenvector. The eigenvectors of the matrix $\mathbf{A}$ are all vectors, $\mathbf{x}_i \neq \mathbf{0}$, which under the transformation $\mathbf{A}$ become multiples of themselves; that is,

$$\mathbf{A}\mathbf{x}_i = \lambda_i\mathbf{x}_i$$

where λ_i 's are constants.

FIGURE 5.32 To be an eigenvector, the transformation $\mathbf{A}\mathbf{x}$ must be collinear with $\mathbf{x}$; thus, in (a), $\mathbf{x}$ is not an eigenvector; in (b), it is.



Eigenvalue. The eigenvalues of the matrix $\mathbf{A}$ are the values of λ_i that satisfy Eq. (5.80) for $\mathbf{x}_i \neq \mathbf{0}$.

To find the eigenvectors, we rearrange Eq. (5.80). Eigenvectors, $\mathbf{x}_i$, satisfy

$$\mathbf{0} = (\lambda_i\mathbf{I} - \mathbf{A})\mathbf{x}_i \quad (5.81)$$

Solving for $\mathbf{x}_i$ by premultiplying both sides by $(\lambda_i\mathbf{I} - \mathbf{A})^{-1}$ yields

$$\mathbf{x}_i = (\lambda_i\mathbf{I} - \mathbf{A})^{-1}\mathbf{0} = \frac{\text{adj}(\lambda_i\mathbf{I} - \mathbf{A})}{\det(\lambda_i\mathbf{I} - \mathbf{A})}\mathbf{0} \quad (5.82)$$

Since $\mathbf{x}_i \neq \mathbf{0}$, a nonzero solution exists if

$$\det(\lambda_i\mathbf{I} - \mathbf{A}) = 0 \quad (5.83)$$

from which λ_i , the eigenvalues, can be found.

□ Diagonal representation

- Equivalent to partial fraction expansion of t.f.
- Use eigenvectors to form a transform matrix $\mathbf{P}$

$$\mathbf{P} = [\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_n]$$

$$\mathbf{A}\mathbf{x}_i = \lambda_i\mathbf{x}_i$$

$$\mathbf{AP} = \mathbf{PD}$$

$$\mathbf{D} = \mathbf{P}^{-1}\mathbf{AP}$$

$$\mathbf{z} = \mathbf{P}^{-1}\mathbf{APz} + \mathbf{P}^{-1}\mathbf{Bu}$$

$$\mathbf{y} = \mathbf{CPz} + \mathbf{Du}$$

Left $\mathbf{D}$ is diagonal matrix with eigenvalues, while right $\mathbf{D}$ is the through-input matrix, often $\mathbf{D} = 0$

Diagonalizing a System in State Space

PROBLEM: Given the system of Eqs. (5.94), find the diagonal system that is similar.

$$\dot{\mathbf{x}} = \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} u \quad (5.94a)$$

$$y = [2 \quad 3] \mathbf{x} \quad (5.94b)$$

$$\mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & -4 \end{bmatrix}$$

$$\mathbf{P}^{-1} \mathbf{B} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3/2 \\ -1/2 \end{bmatrix}$$

$$\mathbf{C} \mathbf{P} = [2 \quad 3] \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = [5 \quad -1]$$

$$\dot{\mathbf{z}} = \begin{bmatrix} -2 & 0 \\ 0 & -4 \end{bmatrix} \mathbf{z} + \begin{bmatrix} 3/2 \\ -1/2 \end{bmatrix} u$$

$$\mathbf{z} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P} \mathbf{z} + \mathbf{P}^{-1} \mathbf{B} u$$

$$y = [5 \quad -1] \mathbf{z}$$

$$y = \mathbf{C} \mathbf{P} \mathbf{z} + \mathbf{D} u$$

$$\begin{aligned} \det(\lambda \mathbf{I} - \mathbf{A}) &= \left| \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \right| \\ &= \begin{vmatrix} \lambda + 3 & -1 \\ -1 & \lambda + 3 \end{vmatrix} \\ &= \lambda^2 + 6\lambda + 8 \end{aligned}$$

$$\mathbf{A} \mathbf{x}_i = \lambda \mathbf{x}_i$$

$$\begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -2 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -4 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} c \\ c \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} c \\ -c \end{bmatrix}$$

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ and } \mathbf{x}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\mathbf{P} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Diagonal state space representation



PROBLEM: For the system represented in state space as follows:

$$\dot{\mathbf{x}} = \begin{bmatrix} 1 & 3 \\ -4 & -6 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 3 \end{bmatrix} u$$
$$y = \begin{bmatrix} 1 & 4 \end{bmatrix} \mathbf{x}$$

convert the system to one where the new state vector, $\mathbf{z}$, is

$$\mathbf{z} = \begin{bmatrix} 3 & -2 \\ 1 & -4 \end{bmatrix} \mathbf{x}$$

ANSWER:

$$\dot{\mathbf{z}} = \begin{bmatrix} 6.5 & -8.5 \\ 9.5 & -11.5 \end{bmatrix} \mathbf{z} + \begin{bmatrix} -3 \\ -11 \end{bmatrix} u$$
$$y = \begin{bmatrix} 0.8 & -1.4 \end{bmatrix} \mathbf{z}$$

PROBLEM: For the original system find the diagonal system that is similar.

ANSWER:

$$\dot{\mathbf{z}} = \begin{bmatrix} -2 & 0 \\ 0 & -3 \end{bmatrix} \mathbf{z} + \begin{bmatrix} 18.39 \\ 20 \end{bmatrix} u$$
$$y = \begin{bmatrix} -2.121 & 2.6 \end{bmatrix} \mathbf{z}$$

Solution of the State Equations



$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$



$$s\mathbf{X}(s) - \mathbf{x}(0) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}\mathbf{U}(s)$$



$$\begin{aligned}\mathbf{X}(s) &= (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(0) + (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{U}(s) \\ &= \frac{\text{adj}(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})} [\mathbf{x}(0) + \mathbf{B}\mathbf{U}(s)]\end{aligned}$$

$$\mathbf{Y}(s) = \mathbf{C}\mathbf{X}(s) + \mathbf{D}\mathbf{U}(s)$$

$$\mathcal{L}^{-1}[(s\mathbf{I} - \mathbf{A})^{-1}] = \mathcal{L}^{-1}\left[\frac{\text{adj}(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})}\right] = \mathbf{\Phi}(t)$$

Solution of the State Equations



$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$



$$s\mathbf{X}(s) - \mathbf{x}(0) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}\mathbf{U}(s)$$



$$\begin{aligned}\mathbf{X}(s) &= (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(0) + (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{U}(s) \\ &= \frac{\text{adj}(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})} [\mathbf{x}(0) + \mathbf{B}\mathbf{U}(s)]\end{aligned}$$

$$\mathbf{Y}(s) = \mathbf{C}\mathbf{X}(s) + \mathbf{D}\mathbf{U}(s)$$

$$\mathcal{L}^{-1}[(s\mathbf{I} - \mathbf{A})^{-1}] = \mathcal{L}^{-1}\left[\frac{\text{adj}(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})}\right] = \mathbf{\Phi}(t)$$

for scalar case

$$X(s) = \frac{x(0)}{s - a} + \frac{b}{s - a}U(s).$$

$$x(t) = e^{at}x(0) + \int_0^t e^{a(t-\tau)}bu(\tau)d\tau.$$

$$\begin{aligned}\mathbf{x}(t) &= e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)d\tau \\ &= \mathbf{\Phi}(t)\mathbf{x}(0) + \int_0^t \mathbf{\Phi}(t - \tau)\mathbf{B}\mathbf{u}(\tau)d\tau\end{aligned}$$

zero-input response

Zero-state response as
convolution integral

The *matrix exponential* function is defined by in a similar Taylor series form

$$e^{\mathbf{A}t} = \exp(\mathbf{A}t) = \mathbf{I} + \mathbf{A}t + \frac{\mathbf{A}^2t^2}{2!} + \cdots + \frac{\mathbf{A}^kt^k}{k!} + \cdots,$$

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

matrix exponentials is a DEFINITION

$$X(t) = e^{tA} = \sum_{i=0}^{\infty} \frac{1}{i!} [tA]^i .$$

Proof:

- Uniqueness of Solutions

$$\mathbf{x}(t) = e^{At}\mathbf{x}(0) + \int_0^t e^{A(t-\tau)}\mathbf{B}\mathbf{u}(\tau)d\tau$$

If we have two solutions $x_1, x_2 : \mathbb{R} \rightarrow \mathbb{R}^{n_x}$, then $y = x_1 - x_2$ satisfies

$$\dot{y}(t) = Ay(t) \quad \text{with} \quad y(0) = 0 .$$

The auxiliary function $v(t) = e^{-At}y(t)$ satisfies

$$\dot{v}(t) = -Ae^{-At}y(t) + e^{-At}Ay(t) = -Ae^{-At}y(t) + Ae^{-At}y(t) = 0$$

$$v(0) = 0 , \quad \implies \quad v(t) = y(t) = 0 \quad \implies \quad x_1 = x_2 .$$

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u$$

matrix exponentials is a DEFINITION

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}u(\tau)d\tau$$

$$X(t) = e^{tA} = \sum_{i=0}^{\infty} \frac{1}{i!} [tA]^i .$$

Proof:

- Verify the ODE

Generalized Leibniz integral rule.

$$\frac{d}{dt} \int_{a(t)}^{b(t)} g(t, \tau) d\tau = g(t, b(t)) \dot{b}(t) - g(t, a(t)) \dot{a}(t) + \int_{a(t)}^{b(t)} g_t(t, \tau) d\tau .$$

$$\begin{aligned} \dot{x}(t) &= \mathbf{A} e^{\mathbf{A}t} x(0) + e^{\mathbf{A}(t-t)} \mathbf{B}u(t) + \int_0^t \mathbf{A} e^{\mathbf{A}(t-\tau)} \mathbf{B}u(\tau) d\tau \\ &= \mathbf{A} \left[e^{\mathbf{A}t} x(0) + \int_0^t e^{\mathbf{A}(t-\tau)} \mathbf{B}u(\tau) d\tau \right] + \mathbf{B}u(t) \\ &= \mathbf{A}x(t) + \mathbf{B}u(t) \end{aligned}$$

Solution of the State Equations



Specially, the solution of *an unforced system*

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t)$$

is found to be

$$\mathbf{x}(t) = \exp(\mathbf{A}t)\mathbf{x}(0)$$

The matrix exponential function describes the unforced response of the system and is called *the fundamental or state transition matrix* $\Phi(t, 0)$.

Thus, the general solution can be written as

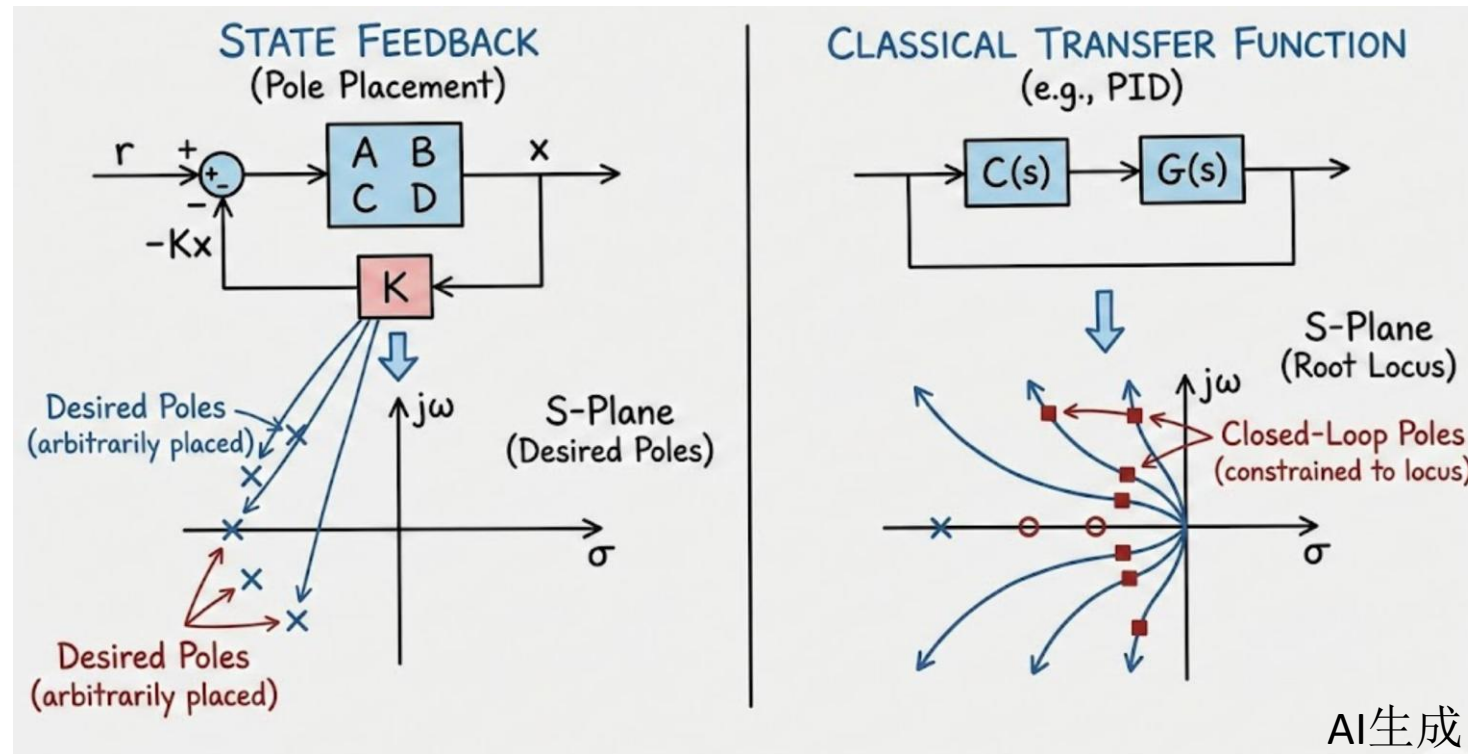
$$\mathbf{x}(t) = \Phi(t)\mathbf{x}(0) + \int_0^t \Phi(t - \tau)\mathbf{B}\mathbf{u}(\tau)d\tau.$$

NOTE, up to now, we are talking about LTI system, for nonlinear or time-varying, there is NO nice general solution form.



State Feedback for Pole Placement

- ❑ Frequency domain methods of design do not allow us to specify all poles in systems of order higher than 2 because they do not allow for a sufficient number of unknown parameters to place all of the closed-loop poles uniquely.
 - One gain to adjust, or compensator pole and zero to select, does not yield a sufficient number of parameters to place all the closed-loop poles at desired locations.
 - Remember, to place n unknown quantities, you need n adjustable parameters.



Comparison of closed-loop pole placement flexibility between State Feedback and Classical approaches.

- Ideally, if the n states of the system are available for feedback, we can introduce n adjustable parameters in the gain matrix K .

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u = \mathbf{A}\mathbf{x} + \mathbf{B}(-\mathbf{K}\mathbf{x} + r) = (\mathbf{A} - \mathbf{B}\mathbf{K})\mathbf{x} + \mathbf{B}r$$

(12.3a)

$$y = \mathbf{C}\mathbf{x}$$

$$\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix}; \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix};$$

$$\mathbf{C} = [c_1 \quad c_2 \quad \cdots \quad c_n]$$

(12.3b)

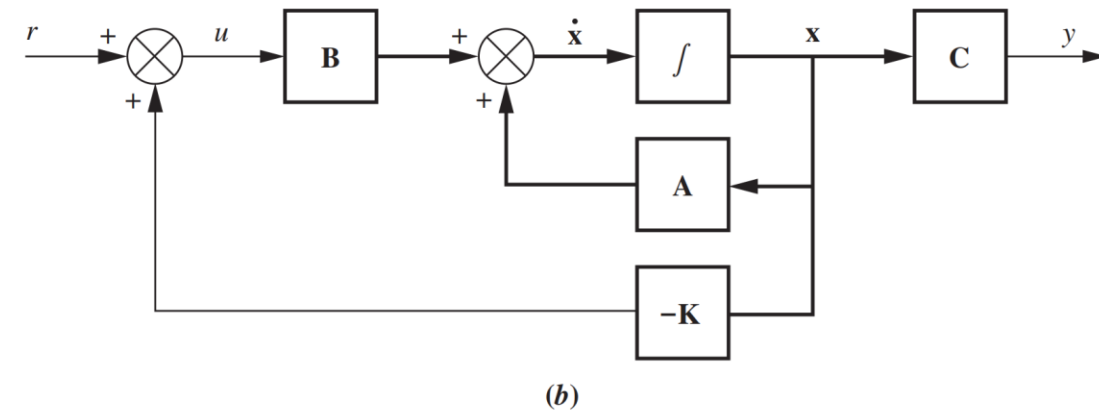
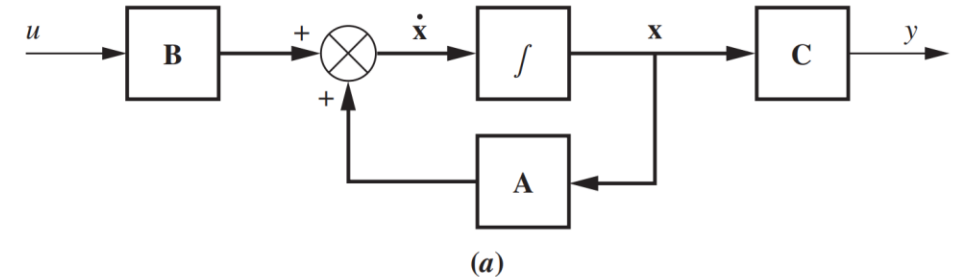


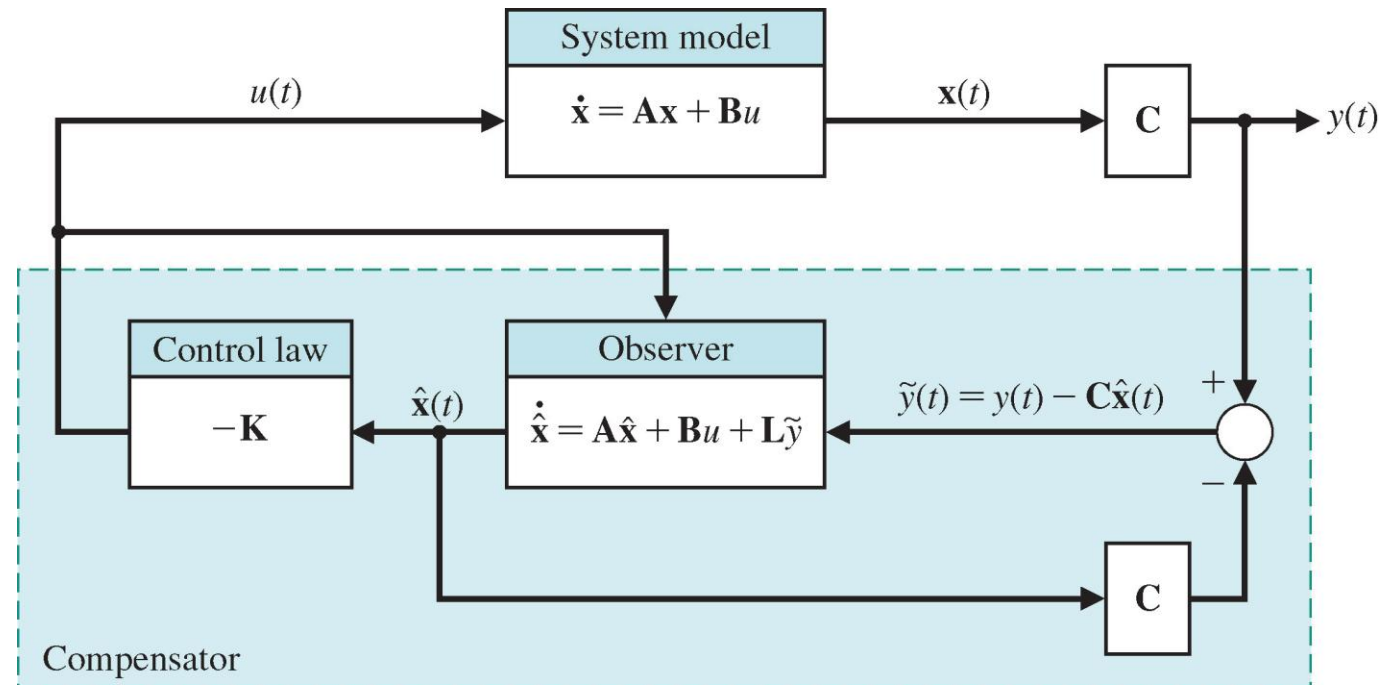
FIGURE 12.2 a. State-space representation of a plant;
b. plant with state-variable feedback

- *State feedback design using only output measurement typically comprises three steps:*

1 we *assume that all the state variables are measurable* and utilize them in a full-state feedback control law.

2 to *construct an observer to estimate the states* that are not directly measured and available as outputs.

3 is to appropriately *connect the observer* to the full-state feedback control law. For linear system, there is a so-called **separation principle** such that the output feedback system just works



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Controllability and Observability

A key question that arises in the design of state variable compensators is whether or not all the poles of the closed-loop system can *be arbitrarily placed* in the complex plane.

The concepts of controllability and observability were introduced by *Kalman* in the 1960s:

if the system is controllable and observable, then we can.

A system is completely controllable if there exists an unconstrained control $u(t)$ that can transfer any initial state $x(t_0)$ to any other desired location $x(t)$ in a finite time, $t_0 \leq t \leq T$.

For the SISO LTI system

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t),$$

we can determine whether the system is controllable by examining the algebraic condition

$$\text{rank}[\mathbf{B} \quad \mathbf{A}\mathbf{B} \quad \mathbf{A}^2\mathbf{B} \quad \dots \quad \mathbf{A}^{n-1}\mathbf{B}] = n.$$

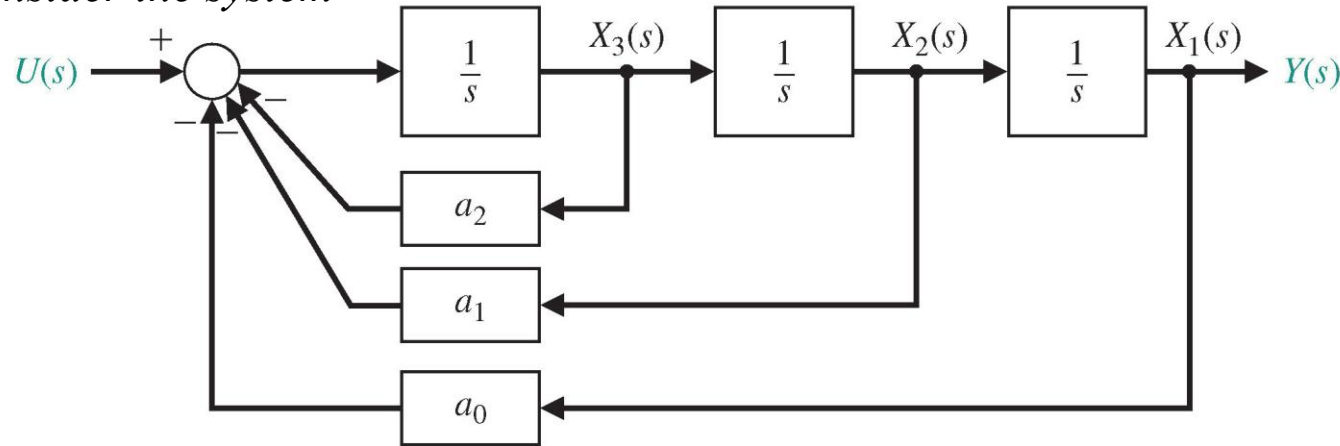
The controllability matrix $\mathbf{P}_c$ is described in terms of $\mathbf{A}$ and $\mathbf{B}$ as

$$\mathbf{P}_c = [\mathbf{B} \quad \mathbf{A}\mathbf{B} \quad \mathbf{A}^2\mathbf{B} \quad \dots \quad \mathbf{A}^{n-1}\mathbf{B}],$$

dimension of the system.

If the determinant of $\mathbf{P}_c$ is nonzero, the system is controllable.

Example: Let us consider the system



Check whether the system is controllable or not?

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t).$$

$$y(t) = [1 \ 0 \ 0] \mathbf{x}(t) + [0] u(t)$$

$$\mathbf{P}_c = [\mathbf{B} \ \mathbf{AB} \ \mathbf{A}^2\mathbf{B}] = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -a_2 \\ 1 & -a_2 & a_2^2 - a_1 \end{bmatrix}.$$

The determinant of $\mathbf{P}_c = -1$, hence this system is controllable.

Observability refers to the *ability to estimate* a state variable.

A system is completely observable if and only if there exists a finite time T such that the initial state $\mathbf{x}(0)$ can be determined from the observation history $\mathbf{y}(t)$ given the control $\mathbf{u}(t)$, $0 \leq t \leq T$.

For the SISO LTI system

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t),$$

This system is completely observable when the determinant of the observability matrix $\mathbf{P}_o$ is nonzero,

$$\mathbf{P}_o = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \vdots \\ \mathbf{CA}^{n-1} \end{bmatrix},$$

Supplementary:

1. Advanced state variable design techniques can handle situations wherein the system is not completely controllable, but where the states (or linear combinations thereof) that *cannot be controlled are inherently stable*. These systems are classified as *stabilizable*.
2. These same techniques can handle cases wherein the system is not completely observable, but where the states (or linear combinations thereof) that *cannot be observed are inherently stable*. These systems are classified as *detectable*.



Controller Design via State Feedback

Full-state Feedback Control Design

Assume that all the states are available for feedback, we have

$$u(t) = -\mathbf{K}\mathbf{x}(t).$$

dimension of
 $\mathbf{K}$?

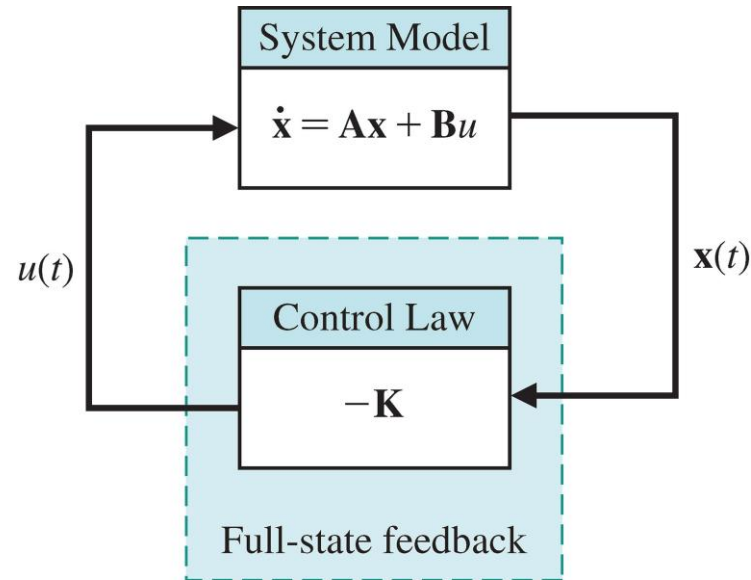


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Determining the gain matrix $\mathbf{K}$ is the objective of the full-state feedback design procedure.



achieve the desired pole locations of the closed-loop system.



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*Theorem : Given the pair (A, B) , we **can always determine $\mathbf{K}$** to place all the system closed-loop poles in the left half-plane if and only if the system is **completely controllable**—that is, if and only if the controllability matrix $\mathbf{P}_c$ is full rank.*

Full-state Feedback Control Design

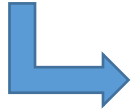


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Assume that all the states are available for feedback, we have

$$u(t) = -\mathbf{K}\mathbf{x}(t).$$

Determining the gain matrix $\mathbf{K}$ is the objective of the full-state feedback design procedure.



achieve the desired pole locations of the closed-loop system.

we find the closed-loop system to be

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) = \mathbf{A}\mathbf{x}(t) - \mathbf{B}\mathbf{K}\mathbf{x}(t) = (\mathbf{A} - \mathbf{B}\mathbf{K})\mathbf{x}(t).$$

The characteristic equation

$$\det(\lambda\mathbf{I} - (\mathbf{A} - \mathbf{B}\mathbf{K})) = 0.$$

If all the roots of the characteristic equation lie in the left half-plane, then the closed-loop system is stable.

In other words, for any initial condition, it follows that

$$\mathbf{x}(t) = e^{(\mathbf{A}-\mathbf{B}\mathbf{K})t}\mathbf{x}(t_0) \rightarrow 0 \quad \text{as } t \rightarrow \infty.$$

*This is known as the **regulator problem**. That is, we want to compute $\mathbf{K}$ so that all initial conditions are driven to zero **in a specified fashion**.*

Full-state Feedback Control Design



Assume that all the states are available for feedback, we have

$$u(t) = -\mathbf{K}\mathbf{x}(t).$$

Determining the gain matrix $\mathbf{K}$ is the objective of the full-state feedback design procedure.

For tracking purpose, addition of a reference input can be written as

$$u(t) = -\mathbf{K}\mathbf{x}(t) + Nr(t),$$

where $r(t)$ is the reference input.

Example: Let us consider the third-order system

$$\frac{d^3y(t)}{dt^3} + 5\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = u(t).$$

We can select the state variables as

$$x_1(t) = y(t),$$

$$x_2(t) = dy(t)/dt,$$

$$x_3(t) = d^2y(t)/dt^2,$$



$$\dot{\mathbf{x}}(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -3 & -5 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t)$$

$$y(t) = [1 \quad 0 \quad 0]\mathbf{x}(t).$$

Controller Design by Matching Coefficients



Example: We design a state feedback controller as

$$u(t) = -\mathbf{K}\mathbf{x}(t), \quad \mathbf{K} = [k_1 \quad k_2 \quad k_3]$$

then the closed-loop system is

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) - \mathbf{B}\mathbf{K}\mathbf{x}(t) = (\mathbf{A} - \mathbf{B}\mathbf{K})\mathbf{x}(t).$$

The state feedback matrix is

$$\mathbf{A} - \mathbf{B}\mathbf{K} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 - k_1 & -3 - k_2 & -5 - k_3 \end{bmatrix},$$

and the characteristic equation is

$$\Delta(\lambda) = \det(\lambda\mathbf{I} - (\mathbf{A} - \mathbf{B}\mathbf{K})) = \lambda^3 + (5 + k_3)\lambda^2 + (3 + k_2)\lambda + (2 + k_1) = 0.$$

If we seek *a rapid response with a low overshoot*, we choose a *desired characteristic* equation such as

$$\Delta(\lambda) = (\lambda^2 + 2\zeta\omega_n\lambda + \omega_n^2)(\lambda + \zeta\omega_n).$$

Controller Design by Matching Coefficients



Example:

$$\Delta(\lambda) = (\lambda^2 + 2\zeta\omega_n\lambda + \omega_n^2)(\lambda + \zeta\omega_n).$$

We choose $\xi = 0.8$ for minimal overshoot and ω_n to meet the settling time requirement

$$T_s = \frac{4}{\zeta\omega_n} = \frac{4}{(0.8)\omega_n} \approx 1. \quad \longrightarrow \quad \omega_n = 6.$$

the desired characteristic equation is

$$(\lambda^2 + 9.6\lambda + 36)(\lambda + 4.8) = \lambda^3 + 14.4\lambda^2 + 82.1\lambda + 172.8.$$

Recall the characteristic equation to be designed

$$\Delta(\lambda) = \det(\lambda\mathbf{I} - (\mathbf{A} - \mathbf{BK})) = \lambda^3 + (5 + k_3)\lambda^2 + (3 + k_2)\lambda + (2 + k_1) = 0.$$

yields the three equations

$$5 + k_3 = 14.4$$

$$3 + k_2 = 82.1$$

$$2 + k_1 = 172.8.$$



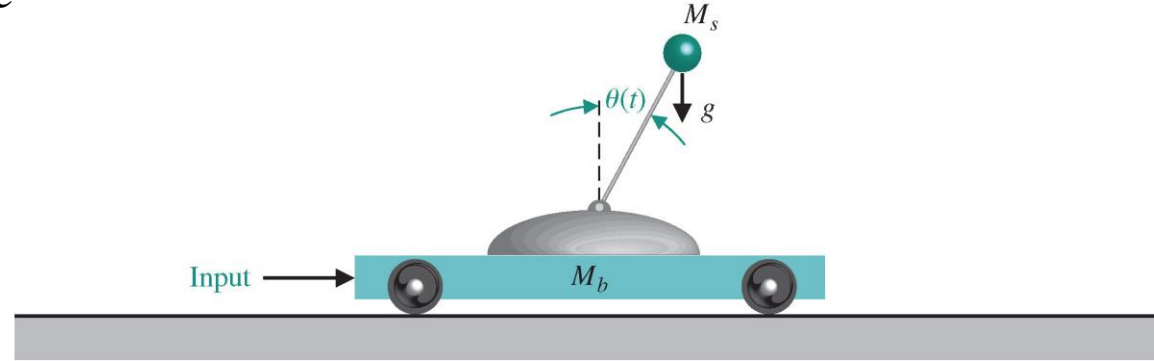
$$K = [170.8, 79.1, 9.4]$$

Controller Design by Matching Coefficients



Example: Inverted pendulum control

Consider the control of an unstable inverted pendulum balanced on a moving cart. Now we tend to design a suitable state variable feedback control system to keep the pendulum staying its unstable position.



*To simplify, we assume that the control input, $u(t)$, is an acceleration signal, **we can focus on the unstable dynamics of the pendulum.***

$$\ddot{\theta}(t) = \frac{g}{l} \theta(t) - \frac{1}{l} u(t).$$

Let the state vector be $(x_1(t), x_2(t)) = (\theta(t), \dot{\theta}(t))$.

$$\frac{d}{dt} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = \begin{bmatrix} 0 & 1 \\ g/l & 0 \end{bmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} + \begin{bmatrix} 0 \\ -1/l \end{bmatrix} u(t).$$

The A matrix has the characteristic equation $\lambda^2 - \frac{g}{l} = 0$ with one root in the right-hand s -plane.

Controller Design by Matching Coefficients



Example: Inverted pendulum control

To stabilize the system, we generate a control signal

$$u(t) = -\mathbf{K}\mathbf{x}(t) = -[k_1 \quad k_2] \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = -k_1 x_1(t) - k_2 x_2(t).$$

Substituting this control signal relationship into the system

$$\begin{pmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{pmatrix} = \begin{bmatrix} 0 & 1 \\ (g + k_1)/l & k_2/l \end{bmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}.$$

Obtaining the characteristic equation, we have

$$\begin{bmatrix} \lambda & -1 \\ -(g + k_1)/l & \lambda - k_2/l \end{bmatrix} = \lambda \left(\lambda - \frac{k_2}{l} \right) - \frac{g + k_1}{l} = \lambda^2 - \left(\frac{k_2}{l} \right) \lambda + \frac{g + k_1}{l}.$$

Thus, for the system to be stable, we require that

$$k_2/l < 0 \text{ and } k_1 > -g.$$

If we wish to achieve a rapid response with modest overshoot, we select

$$\omega_n = 10 \text{ and } \zeta = 0.8.$$

Then we require

$$\frac{k_2}{l} = -16 \quad \text{and} \quad \frac{k_1 + g}{l} = 100.$$

Controller Design by Matching Coefficients



Tips for making our life easier:

1. For a SISO LTI, Given the desired characteristic equation

$$q(\lambda) = \lambda^n + \alpha_{n-1}\lambda^{n-1} + \cdots + \alpha_0,$$

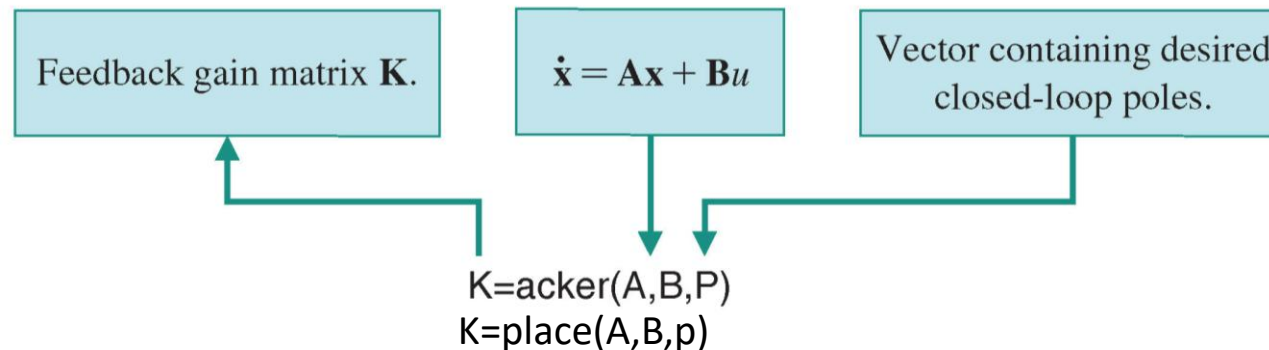
Then *Ackermann's formula* is useful for determining the state variable feedback matrix:

$$\mathbf{K} = [0 \ 0 \ \dots \ 0 \ 1] \mathbf{P}_c^{-1} q(\mathbf{A}),$$

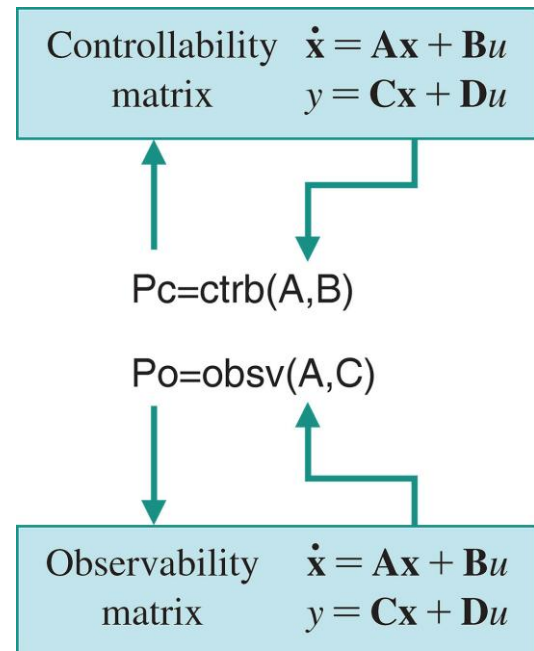
where P_c is the controllability matrix and

$$q(\mathbf{A}) = \mathbf{A}^n + \alpha_{n-1}\mathbf{A}^{n-1} + \cdots + \alpha_1\mathbf{A} + \alpha_0\mathbf{I},$$

2. Matlab Code:



Invertible
Matrix



- We find the phase variable version of a controllable system, and design the gain matrix.
 - It can be proved that the transformation matrix $\mathbf{P}$ is related to the two controllability matrices:

$$\mathbf{P} = \mathbf{C}_{\mathbf{Mz}} \mathbf{C}_{\mathbf{Mx}}^{-1}$$

- The gain matrix for the original system can be derived using the transformation matrix $\mathbf{P}$.

$$\mathbf{K}_z = \mathbf{K}_x \mathbf{P}^{-1}$$

$$\dot{\mathbf{z}} = \mathbf{A}_z \mathbf{z} + \mathbf{B}_z u = \begin{bmatrix} -5 & 1 & 0 \\ 0 & -2 & 1 \\ 0 & 0 & -1 \end{bmatrix} \mathbf{z} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = \mathbf{C}_z \mathbf{z} = [-1 \quad 1 \quad 0] \mathbf{z}$$

Open loop poles

$$\dot{\mathbf{x}} = \mathbf{A}_x \mathbf{x} + \mathbf{B}_x u = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -10 & -17 & -8 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = [4 \quad 1 \quad 0] \mathbf{x}$$

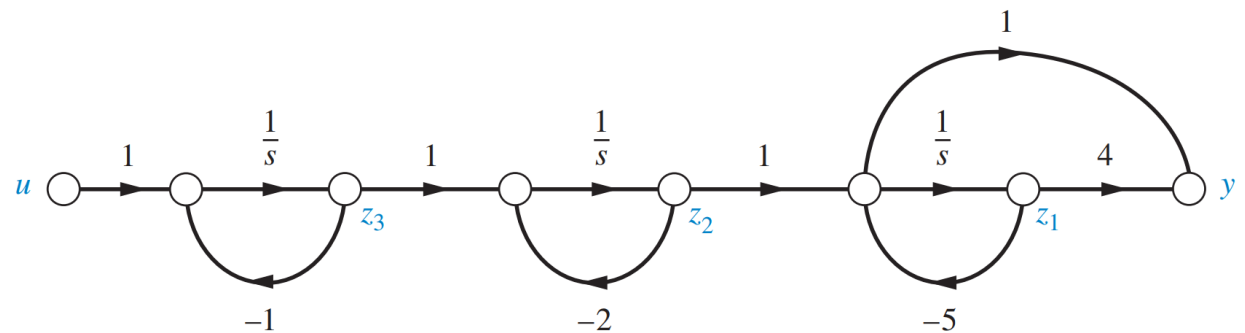
Characteristic equation coefficients

FIGURE 12.9 Signal-flow graph for plant of Example 12.4

PROBLEM: Design a state-variable feedback controller to yield a 20.8% overshoot and a settling time of 4 seconds for a plant,

$$G(s) = \frac{(s + 4)}{(s + 1)(s + 2)(s + 5)} \quad (12.43)$$

that is represented in cascade form as shown in Figure 12.9.



Controller Design by Transformation



PROBLEM: Design a linear state-feedback controller to yield 20% overshoot and a settling time of 2 seconds for a plant

$$G(s) = \frac{(s + 6)}{(s + 9)(s + 8)(s + 7)}$$

that is represented in state space in cascade form by

$$\dot{\mathbf{z}} = \mathbf{A}\mathbf{z} + \mathbf{B}u = \begin{bmatrix} -7 & 1 & 0 \\ 0 & -8 & 1 \\ 0 & 0 & -9 \end{bmatrix} \mathbf{z} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$
$$y = \mathbf{C}\mathbf{z} = \begin{bmatrix} -1 & 1 & 0 \end{bmatrix} \mathbf{z}$$

ANSWER:





Partially Controllable System

- Given a single-input continuous-time linear time-invariant controlled system, determine whether there is a state feedback matrix K so that the closed-loop eigenvalues are configured to the following positions.

$$\dot{x} = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} u$$

(a) $\lambda_1^* = -2, \lambda_2^* = -2, \lambda_3^* = -2, \lambda_4^* = -2$

(b) $\lambda_1^* = -3, \lambda_2^* = -3, \lambda_3^* = -3, \lambda_4^* = -2$

(c) $\lambda_1^* = -3, \lambda_2^* = -4, \lambda_3^* = -3, \lambda_4^* = -3$

□ Write out the differential equations for x_3 and x_4 :

- $\dot{x}_3 = -2x_3 + 1u$
- $\dot{x}_4 = -2x_4 + 1u$
- Define a new state denoted by $\Delta x = x_3 - x_4$,
- which is governed by $\frac{d}{dt}\Delta x = -2\Delta x$
- The eigenvalue of the new state Δx cannot be modified by u .
 - This is what essentially the concept of uncontrollable state.

□ For the first block, we can use the idea of back-stepping:

- To control x_1 or do pole placement, we can use x_2 as virtual input v .
- That is, $x_2 = \frac{1}{s}(-2x_2 + u) = v$
- If we want to place the first eigenvalue to -12 , then we can design $v = -14x_1$,
- which gives $\dot{x}_1 = 2x_1 + v = -12x_1$
- Even though the DoF of $x_2 = -14x_1$ has been used, its eigenvalue is not yet been determined.
- We can design $u = -14x_2$ to place the eigenvalue of x_2 to a new location.
- $\dot{x}_2 = 2x_2 + 1u = -12x_2$.
- The derivative of x_2 is related to its eigenvalue. The value of $x_2 = v$ is used to place the eigenvalue of x_1 .

$$\dot{x} = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} u$$

$$(a) \lambda_1^* = -2, \lambda_2^* = -2, \lambda_3^* = -2, \lambda_4^* = -2$$

$$(b) \lambda_1^* = -3, \lambda_2^* = -3, \lambda_3^* = -3, \lambda_4^* = -2$$

$$(c) \lambda_1^* = -3, \lambda_2^* = -4, \lambda_3^* = -3, \lambda_4^* = -3$$

- ❑ Further consider this state space model and do what we have done again.
- ❑ Write out the differential equation for x_3 and x_4 :
 - $\dot{x}_3 = -2x_3 + 1u$
 - $\dot{x}_4 = -2x_4 + 2u$
 - Define a new state denoted by $\Delta x = 2x_3 - x_4$,
 - which is governed by $\frac{d}{dt}\Delta x = -4x_3 + 2x_4 = -2\Delta x$
 - The eigenvalue of the new state Δx cannot be modified by u .
 - This is again what essentially the concept of uncontrollable state.
- ❑ For the first block, let's also cancel the input u by defining a new delta state.
 - $\frac{d}{dt}\Delta x = \frac{d}{dt}(x_1 - x_2) = 3x_1 + x_2 - 3x_2 = -3\Delta x + x_2$
 - Here, x_2 is affected by u and can be treated as a virtual input.

$$\dot{x} = \begin{bmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \end{bmatrix} u$$

- Finally, let's also check the eigenvalues of 3 and -2 they are not closed dynamics (closed dynamics cannot be affected by input u)
- $\dot{x}_2 = 3x_2 + 1u$
 - $\dot{x}_3 = -2x_3 + 1u$
 - $\frac{d}{dt}\Delta x = 3x_2 + 2x_3 = 3\Delta x + 5x_3$
 - Now the state x_3 can be viewed as a virtual input that is affected by input u .

$$\dot{x} = \begin{bmatrix} 3 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \end{bmatrix} u$$

- ❑ The reason why a matrix's determinant is equal to the product of all its eigenvalues.

◆ 对于约当标准型矩阵 (Jordan Canonical Form, JCF), 其行列式的计算非常简单:

约当标准型矩阵的行列式等于其主对角线上所有元素的乘积。

如果一个约当标准型矩阵 J 的主对角线元素为 $\lambda_1, \lambda_2, \dots, \lambda_n$ (这些就是矩阵的特征值), 那么:

$$\det(J) = \lambda_1 \cdot \lambda_2 \cdot \dots \cdot \lambda_n$$

为什么可以这样计算?

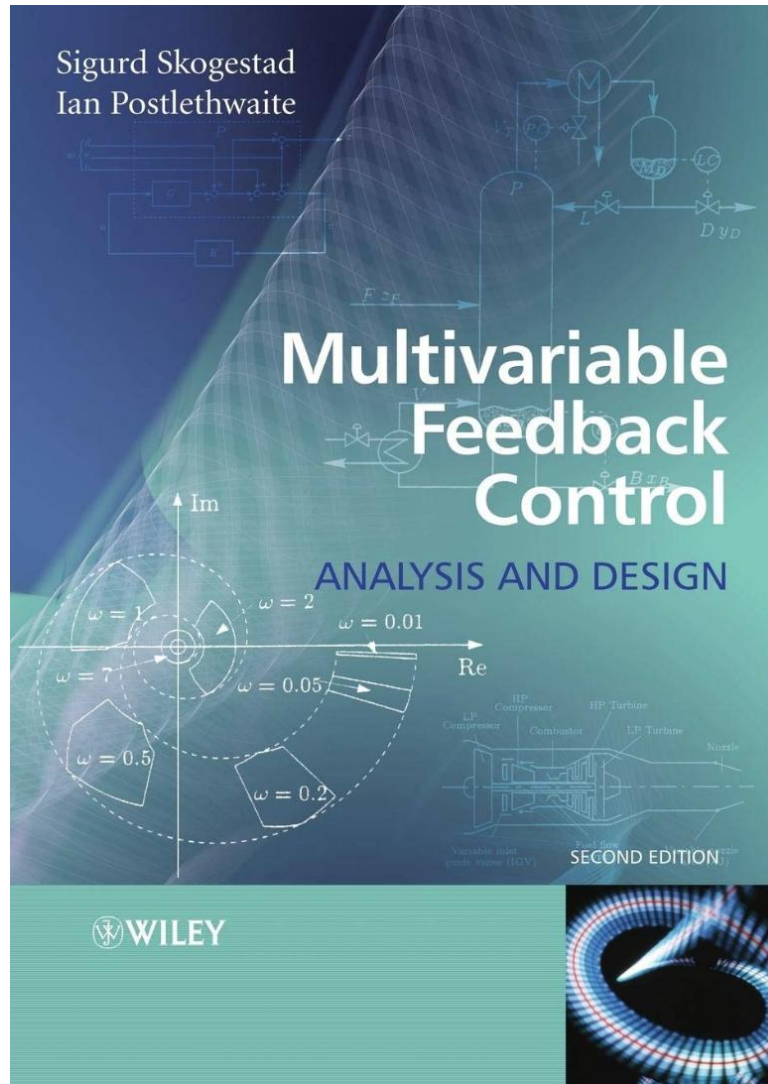
这个结论基于两个基本的线性代数性质:

1. **三角矩阵的行列式性质:** 约当标准型矩阵是一个上三角矩阵 (Upper Triangular Matrix)。它的非零元素只出现在主对角线以及主对角线上方的一条次对角线上, 主对角线下方全为零。
 - 对于任何上三角矩阵 (或下三角矩阵), 其行列式都等于主对角线上所有元素的乘积。
 - 直观理解: 在计算行列式时, 可以使用拉普拉斯展开 (Laplace Expansion)。对于上三角矩阵, 如果我们沿着第一列展开, 只需要考虑 a_{11} 这一项, 因为它乘以的代数余子式仍然是一个上三角矩阵。以此类推, 最终结果就是对角线元素的乘积。
2. **相似矩阵的行列式性质:** 任何方阵 A 都相似于其约当标准型 J 。这意味着存在一个可逆矩阵 P , 使得 $A = PJP^{-1}$ 。
 - 根据行列式的乘法性质: $\det(AB) = \det(A) \det(B)$ 。
 - 因此, $\det(A) = \det(PJP^{-1}) = \det(P) \det(J) \det(P^{-1})$ 。
 - 由于 $\det(P^{-1}) = 1 / \det(P)$, 它们可以相互抵消。
 - 结论: $\det(A) = \det(J)$ 。

总结: 因为约当标准型 J 是一个上三角矩阵, 所以它的行列式等于对角线元素 (特征值) 的乘积。又因为原矩阵 A 与 J 相似, 所以它们的行列式相等。这就是为什么矩阵 A 的行列式也等于其所有特征值乘积的原因。



The Bible for Modern (State Space) Control System



4.2 State controllability and state observability

Definition 4.1 State controllability *The dynamical system $\dot{x} = Ax + Bu$ or the pair (A, B) is said to be state controllable if, for any initial state $x(0) = x_0$, any time $t_1 > 0$ and any final state x_1 , there exists an input $u(t)$ such that $x(t_1) = x_1$. Otherwise the system is said to be state uncontrollable.*

From (4.7) one can verify that a particular input which achieves $x(t_1) = x_1$ is

$$u(t) = -B^T e^{A^T(t_1-t)} W_c(t_1)^{-1} (e^{At_1} x_0 - x_1) \quad (4.40)$$

where $W_c(t)$ is the Gramian matrix at time t ,

$$W_c(t) \triangleq \int_0^t e^{A\tau} B B^T e^{A^T \tau} d\tau \quad (4.41)$$

There are many ways to check whether a system is state controllable. First, we have that (A, B) is state controllable if and only if the controllability matrix

$$C \triangleq [B \quad AB \quad A^2B \quad \dots \quad A^{n-1}B] \quad (4.42)$$

has rank n (full row rank). Here n is the number of states.



Observer

$$u(t) = -\mathbf{K}\mathbf{x}(t).$$

Next big question,

- what if the state $\mathbf{x}(t)$ is not available?
- Especially when $\mathbf{x}(t)$ does not have a good physical meaning.
- Or when you can afford an expensive sensor?

Fortunately, *if the system is completely observable with a given set of outputs*, then it is possible to determine (or to estimate) the states that are not directly measured (or observed).

According to *Luenberger*, the full-state observer for the system

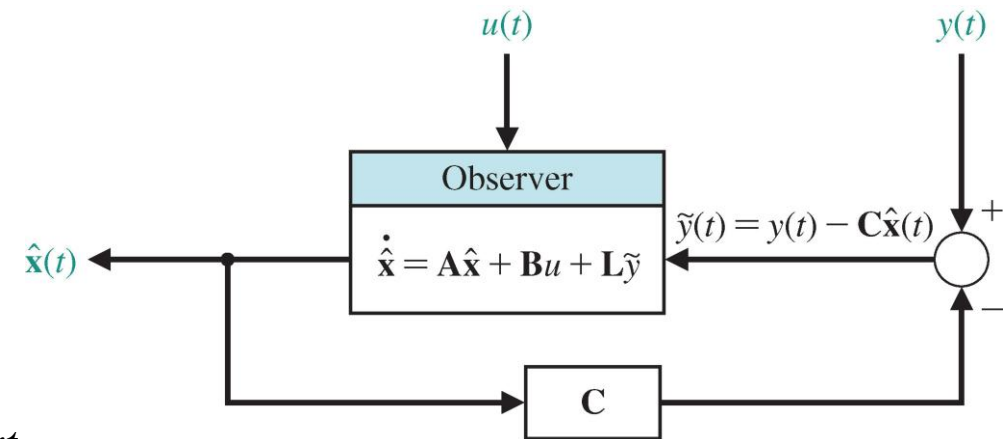
$$\begin{aligned}\dot{\mathbf{x}}(t) &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) \\ y(t) &= \mathbf{C}\mathbf{x}(t)\end{aligned}$$

is given by

$$\dot{\hat{\mathbf{x}}}(t) = \mathbf{A}\hat{\mathbf{x}}(t) + \mathbf{B}u(t) + \mathbf{L}(y(t) - \mathbf{C}\hat{\mathbf{x}}(t))$$

where

- $\hat{\mathbf{x}}(t)$ is the so-called estimates of the state $\mathbf{x}(t)$
- *matrix $\mathbf{L}$* is the observer gain matrix and is to be determined as part of the observer design procedure.



$$\dot{\hat{\mathbf{x}}}(t) = \mathbf{A}\hat{\mathbf{x}}(t) + \mathbf{B}u(t) + \mathbf{L}(y(t) - \mathbf{C}\hat{\mathbf{x}}(t))$$

$$\hat{x}(t_0) = \hat{x}_0$$

in general, not equal x_0

The goal of the observer is to provide an estimate $\hat{x}(t) \rightarrow x(t)$ as $t \rightarrow \infty$.

Define the observer estimation error as

$$\mathbf{e}(t) = \mathbf{x}(t) - \hat{\mathbf{x}}(t).$$

The observer design should produce an observer with the property that

$$e(t) \rightarrow 0 \text{ as } t \rightarrow \infty$$

Taking the time-derivative of the estimation error

$$\begin{aligned}\dot{\mathbf{e}}(t) &= \dot{\mathbf{x}}(t) - \dot{\hat{\mathbf{x}}}(t) \\ &= \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) - \mathbf{A}\hat{\mathbf{x}}(t) - \mathbf{B}u(t) - \mathbf{L}(y(t) - \mathbf{C}\hat{\mathbf{x}}(t)) \\ &= (\mathbf{A} - \mathbf{LC})\mathbf{e}(t).\end{aligned}$$

Conclusion: We can achieve our goal if the characteristic equation

$$\det(\lambda\mathbf{I} - (\mathbf{A} - \mathbf{LC})) = 0$$

has all its roots in the left half-plane.

Example: Consider the second-order system

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$y(t) = [1 \quad 0] \mathbf{x}(t).$$

In this example, we can only directly observe the state $y(t) = x_1(t)$.

checking the system observability

$$\mathbf{P}_o = \begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}.$$

The full-state observer for the system

$$\dot{\hat{\mathbf{x}}}(t) = \mathbf{A}\hat{\mathbf{x}}(t) + \mathbf{B}u(t) + \mathbf{L}(y(t) - \mathbf{C}\hat{\mathbf{x}}(t))$$

where

$$\mathbf{L} = [L_1 \quad L_2]^T.$$

Then the characteristic equation of the estimation error yields

$$\det(\lambda \mathbf{I} - (\mathbf{A} - \mathbf{LC})) = \lambda^2 + (L_1 - 6)\lambda - 4(L_1 - 2) + 3(L_2 + 1),$$

Observer Design

Suppose that the desired characteristic equation is given by

$$\Delta_d(\lambda) = \lambda^2 + 2\zeta\omega_n\lambda + \omega_n^2.$$

We can select $\xi = 0.8$ and $\omega_n = 10$, resulting in an expected settling time of less than 0.5 second.

Equating the coefficients

$$L_1 - 6 = 16$$

$$-4(L_1 - 2) + 3(L_2 + 1) = 100$$

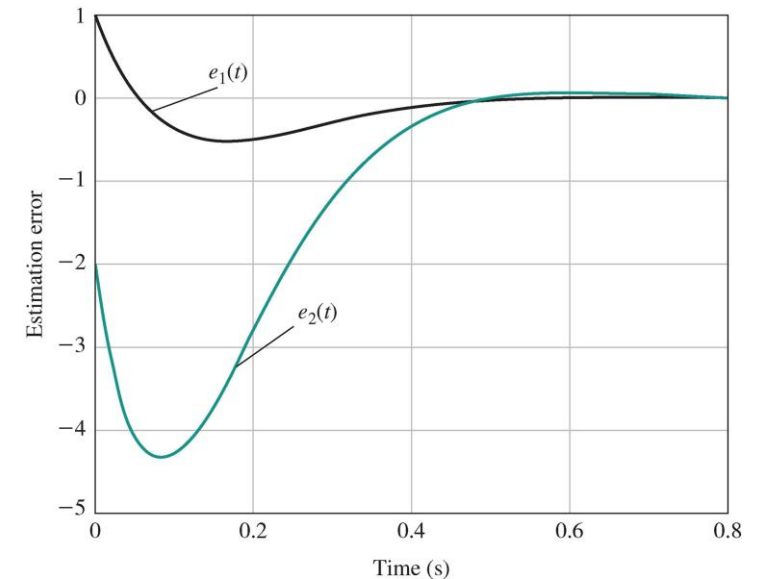
which, when solved, produces

$$\mathbf{L} = \begin{bmatrix} L_1 \\ L_2 \end{bmatrix} = \begin{bmatrix} 22 \\ 59 \end{bmatrix}.$$

The observer is thus given by

$$\dot{\hat{\mathbf{x}}}(t) = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix} \hat{\mathbf{x}}(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t) + \begin{bmatrix} 22 \\ 59 \end{bmatrix} (y(t) - \hat{x}_1(t)).$$

what does this
“settling” mean?



Note:

1. *Given the desired observer characteristic equation*

$$p(\lambda) = \lambda^n + \beta_{n-1}\lambda^{n-1} + \dots + \beta_1\lambda + \beta_0.$$

Ackermann's formula can also be employed to place the roots of the observer

$$\mathbf{L} = p(\mathbf{A})\mathbf{P}_o^{-1}[0 \dots 0 \ 1]^T,$$

where P_o is the observability matrix

$$p(\mathbf{A}) = \mathbf{A}^n + \beta_{n-1}\mathbf{A}^{n-1} + \dots + \beta_1\mathbf{A} + \beta_0\mathbf{I}.$$

2. *Up to now, the effectiveness of the observer has **NOTHING** to do with the control input and it will **NOT** alter the behaviour of the system*

Observer based Feedback Control



Recall

$$u(t) = -\mathbf{K}\mathbf{x}(t).$$

*It seems reasonable that we can employ the state estimate in the feedback control law **in place of** $\mathbf{x}(t)$.*

$$u(t) = -\mathbf{K}\hat{\mathbf{x}}(t).$$

We need to verify that, using the estimate still retain the stability of the closed-loop system.

Proof: Substituting the observer-based feedback law into the system model yields

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) = \mathbf{A}\mathbf{x}(t) - \mathbf{B}\mathbf{K}\hat{\mathbf{x}}(t),$$

and with $\hat{\mathbf{x}}(t) = \mathbf{x}(t) - \mathbf{e}(t)$ we obtain

$$\dot{\mathbf{x}}(t) = (\mathbf{A} - \mathbf{B}\mathbf{K})\mathbf{x}(t) + \mathbf{B}\mathbf{K}\mathbf{e}(t).$$

Recall the the estimation error

$$\dot{\mathbf{e}}(t) = \dot{\mathbf{x}}(t) - \dot{\hat{\mathbf{x}}}(t) = (\mathbf{A} - \mathbf{L}\mathbf{C})\mathbf{e}(t).$$

$$\begin{pmatrix} \dot{\mathbf{x}}(t) \\ \dot{\mathbf{e}}(t) \end{pmatrix} = \begin{bmatrix} \mathbf{A} - \mathbf{B}\mathbf{K} & \mathbf{B}\mathbf{K} \\ \mathbf{0} & \mathbf{A} - \mathbf{L}\mathbf{C} \end{bmatrix} \begin{pmatrix} \mathbf{x}(t) \\ \mathbf{e}(t) \end{pmatrix}.$$

$$\Rightarrow \Delta(\lambda) = \det(\lambda\mathbf{I} - (\mathbf{A} - \mathbf{B}\mathbf{K}))\det(\lambda\mathbf{I} - (\mathbf{A} - \mathbf{L}\mathbf{C}))$$

So if $\mathbf{A}-\mathbf{B}\mathbf{K}$ and $\mathbf{A}-\mathbf{L}\mathbf{C}$ are both Hurwitz, then the overall system is stable.

Observer based Feedback Control



*The fact that the full-state feedback law and the observer can be designed independently is an illustration of **the separation principle**.*

The design procedure is summarized as follows:

1. Determine $\mathbf{K}$ such that $\det(\lambda\mathbf{I} - (\mathbf{A} - \mathbf{BK})) = 0$ has roots in the left half-plane and place the poles appropriately to meet the control system design specifications. The ability to place the poles arbitrarily in the complex plane is guaranteed if the system is completely controllable.
2. Determine $\mathbf{L}$ such that $\det(\lambda\mathbf{I} - (\mathbf{A} - \mathbf{LC})) = 0$ has roots in the left half-plane and place the poles to achieve acceptable observer performance. The ability to place the observer poles arbitrarily in the complex plane is guaranteed if the system is completely observable.
3. Connect the observer to the full-state feedback law using

$$u(t) = -\mathbf{K}\hat{\mathbf{x}}(t).$$

Compensator Transfer Function.

$$U(s) = [-\mathbf{K}(s\mathbf{I} - (\mathbf{A} - \mathbf{BK} - \mathbf{LC}))^{-1}\mathbf{L}]Y(s).$$

*This controller is also commonly referred to as a **stabilizing controller**.*

Note that, even though $A - BK$ is stable and $A - LC$ is stable, it does not necessarily follow that $A - BK - LC$ is stable.

Observer based Feedback Control

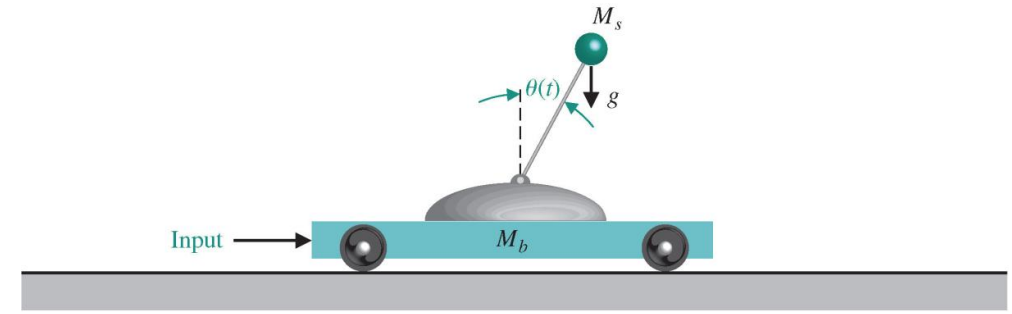


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Example: Compensator design for the inverted pendulum

$$\dot{\mathbf{x}}(t) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{-mg}{M} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{g}{l} & 0 \end{bmatrix} \mathbf{x}(t) + \begin{bmatrix} 0 \\ \frac{1}{M} \\ 0 \\ \frac{-1}{Ml} \end{bmatrix} u(t),$$

$$y(t) = [1 \ 0 \ 0 \ 0] \mathbf{x}(t).$$



Let the system parameters be

$$l = 0.098 \text{ m}, g = 9.8 \text{ m/s}^2, m = 0.825 \text{ kg} \quad M = 8.085 \text{ kg}.$$

Checking controllability yields the controllability matrix

$$\mathbf{P}_c = \begin{bmatrix} 0 & 0.1237 & 0 & 1.2621 \\ 0.1237 & 0 & 1.2621 & 0 \\ 0 & -1.2621 & 0 & -126.21 \\ -1.2621 & 0 & -126.21 & 0 \end{bmatrix}.$$

$$\mathbf{P}_o = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

Observer based Feedback Control



Step 1: Design the Full-State Feedback Control Law

The open-loop system poles are located at

$$\lambda = 0, 0, -10, \text{ and } 10,$$

hence the open-loop system is unstable (there is a pole in the right half-plane).

$$q(\lambda) = (\lambda^2 + 2\zeta\omega_n\lambda + \omega_n^2)(\lambda^2 + a\lambda + b),$$

To obtain a settling time less than 10 seconds with low overshoot, we can select

$$(\zeta, \omega_n) = (0.8, 0.5).$$

*Then, we choose a **separation factor of 20** between the dominant poles and the remaining poles, from which it follows that*

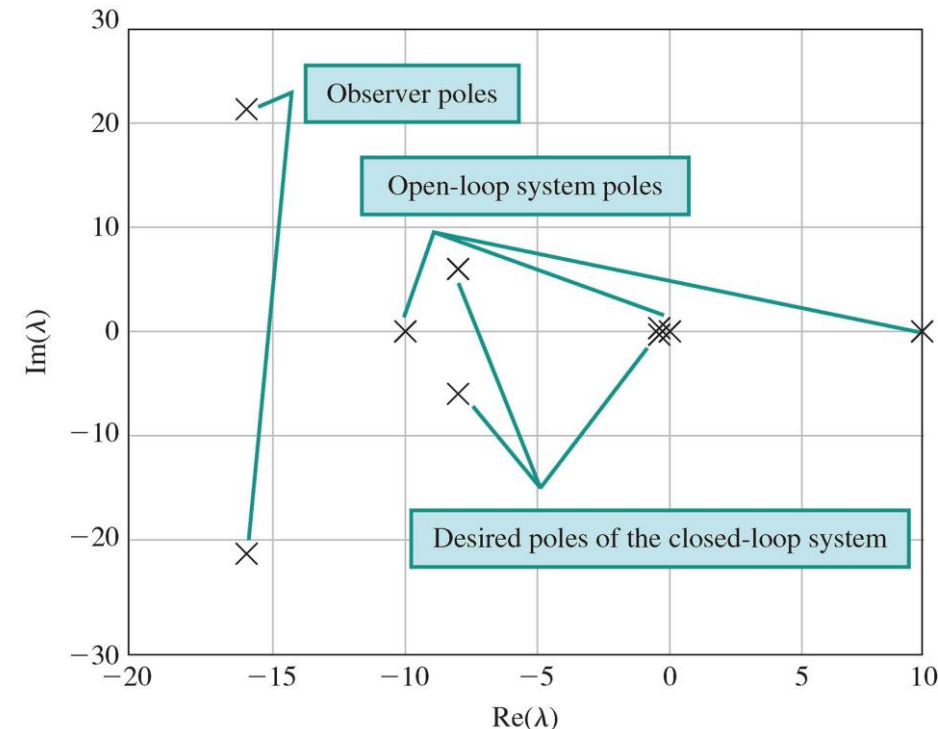
$$(a, b) = (16, 100)$$

The desired roots are

$$\det(\lambda \mathbf{I} - (\mathbf{A} - \mathbf{BK})) = (\lambda + 8 \pm j6)(\lambda + 0.4 \pm j0.3).$$

Using Ackermann's formula yields the feedback gain matrix

$$\mathbf{K} = [-2.2509 \quad -7.5631 \quad -169.0265 \quad -14.0523].$$



Observer based Feedback Control



Step 2: Observer Design

The desired observer characteristic equation is selected to be of the form

$$p(\lambda) = (\lambda^2 + c_1\lambda + c_2)^2$$

where

$$c_1 = 32 \text{ and } c_2 = 711.11.$$

These values should produce a response to an initial state estimation error that settles in less than 0.5 second with minimal percent overshoot.

Using Ackermann's formula, we have

$$\mathbf{L} = \begin{bmatrix} 64.0 \\ 2546.22 \\ -5.1911\text{E}04 \\ -7.6030\text{E}05 \end{bmatrix}.$$

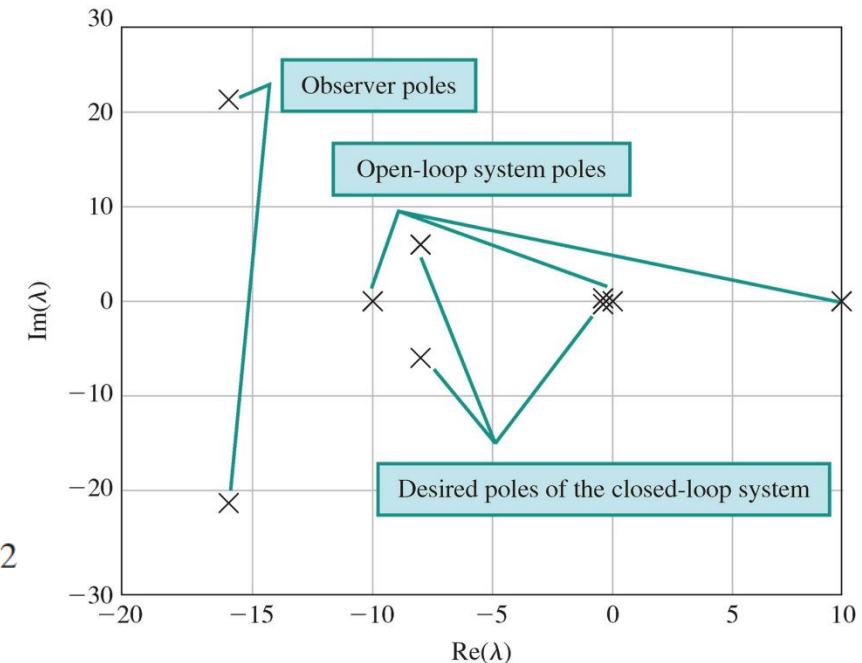
achieves the desired observer pole locations

$$\det(\lambda\mathbf{I} - (\mathbf{A} - \mathbf{LC})) = ((\bar{\lambda} + 16 + j21.3)(\lambda + 16 - j21.3))^2$$

The goal is to achieve an accurate estimate as fast as possible without resulting in too large a gain matrix $\mathbf{L}$.

How large is too large depends on the problem under consideration.

The *trade-off* between the time required to obtain accurate observer performance and the amount of noise amplification is a primary design issue



Observer based Feedback Control

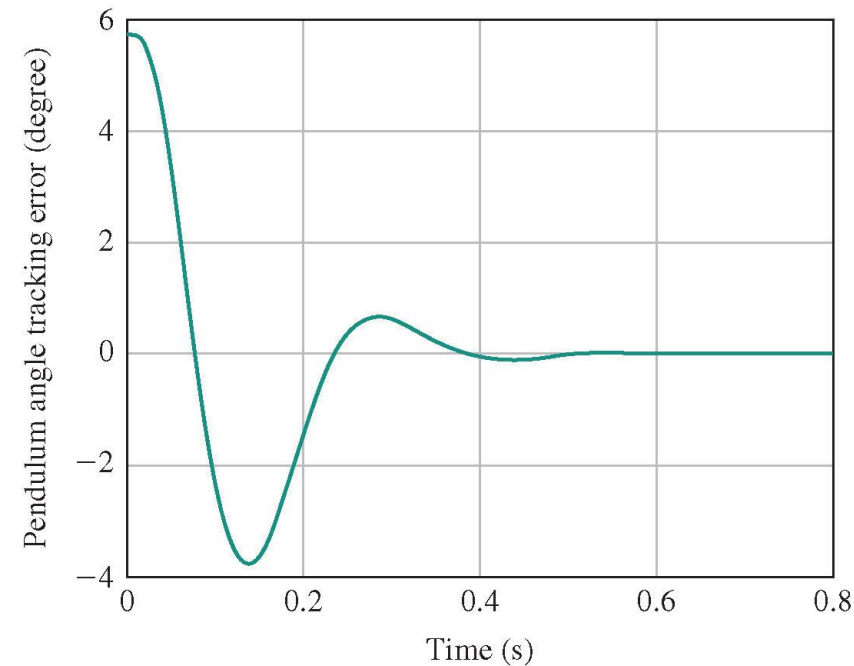
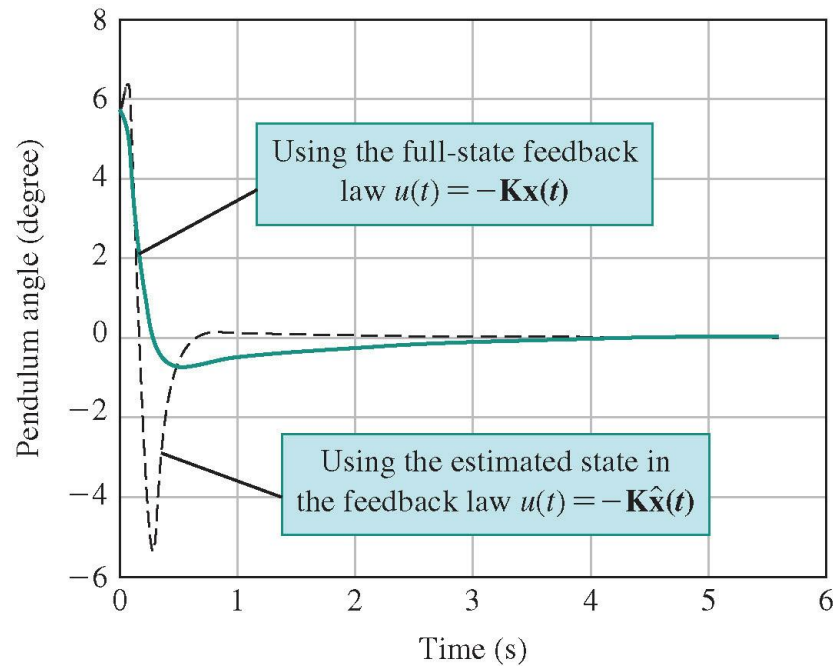


Step 3: Compensator Design

The final step in the design is to connect the observer to the full-state feedback control law via

$$u(t) = -\mathbf{K}\hat{\mathbf{x}}(t)$$

This stabilize the closed-loop system, however, we should *not expect* the closed-loop performance to be as good when using the state estimate from the observer.



Observer Design Example



- Given a continuous-time linear time-invariant system, design a Luenberger observer such that the roots of the characteristic equation of the dynamic of the observation error are all located at -3, -3, -4.

$$\dot{x} = \begin{bmatrix} -1 & -2 & -2 \\ 0 & -1 & 1 \\ 1 & 0 & -1 \end{bmatrix} x + \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} u$$
$$y = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} x$$



Reference Input

Reference Input



We referred to the previous design of state variable feedback stabilizing compensators without reference inputs (i.e., $r(t) = 0$) as **regulators**, however, **command following (trajectory tracking)** is also an important aspect of feedback design

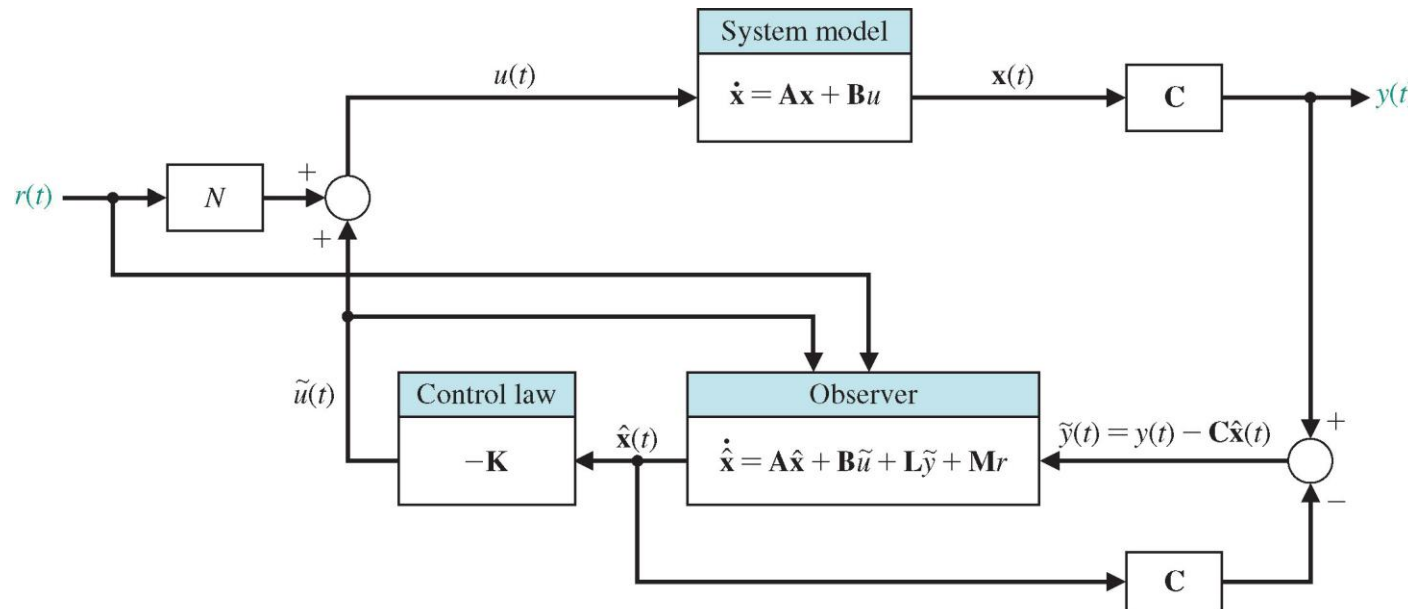
Let's consider *how we can introduce a reference signal* into the state variable feedback compensator.

$$\dot{\hat{\mathbf{x}}}(t) = \mathbf{A}\hat{\mathbf{x}}(t) + \mathbf{B}\tilde{u}(t) + \mathbf{L}\tilde{y}(t) + \mathbf{M}r(t)$$

$$u(t) = \tilde{u}(t) + Nr(t) = -\mathbf{K}\hat{\mathbf{x}}(t) + Nr(t),$$

where

$$\tilde{y}(t) = y(t) - \mathbf{C}\hat{\mathbf{x}}(t)$$



The compensator key design parameters required to implement the command tracking of the reference input are **M and N** .

Case 1, we select M and N so that the estimation error $e(t)$ is independent of the reference input $r(t)$

the estimation error

$$\begin{aligned}\dot{\mathbf{e}}(t) &= \dot{\mathbf{x}}(t) - \dot{\hat{\mathbf{x}}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t) - \mathbf{A}\hat{\mathbf{x}}(t) - \mathbf{B}\tilde{u}(t) - \mathbf{L}\tilde{y}(t) - \mathbf{M}r(t), \\ &= (\mathbf{A} - \mathbf{L}\mathbf{C})\mathbf{e}(t) + (\mathbf{B}N - \mathbf{M})r(t).\end{aligned}$$

Suppose that we select

$$\mathbf{M} = \mathbf{B}N.$$

Then the corresponding estimation error is given by

$$\dot{\mathbf{e}}(t) = (\mathbf{A} - \mathbf{L}\mathbf{C})\mathbf{e}(t).$$

In this case, the estimation error is independent of the reference input, and the remaining task is *to determine a suitable value of N* .

With $M = BN$, we find that the compensator is given by

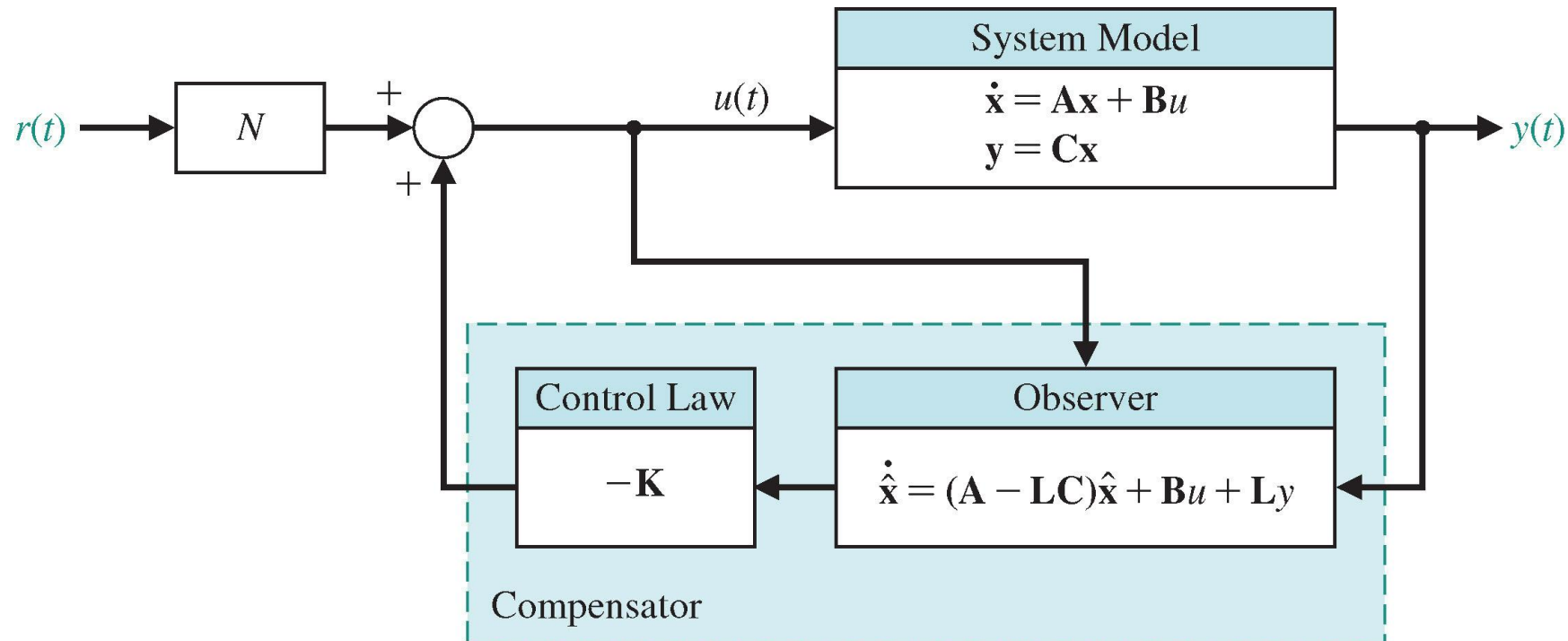
$$\begin{aligned}\dot{\hat{\mathbf{x}}}(t) &= \mathbf{A}\hat{\mathbf{x}}(t) + \mathbf{B}u(t) + \mathbf{L}\tilde{y}(t) \\ u(t) &= -\mathbf{K}\hat{\mathbf{x}}(t) + Nr(t).\end{aligned}$$

Reference Input



Case 1, we select M and N so that the estimation error $e(t)$ is independent of the reference input $r(t)$

$$\begin{aligned}\dot{\hat{\mathbf{x}}}(t) &= \mathbf{A}\hat{\mathbf{x}}(t) + \mathbf{B}u(t) + \mathbf{L}\tilde{y}(t) \\ u(t) &= -\mathbf{K}\hat{\mathbf{x}}(t) + Nr(t).\end{aligned}$$



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the compensator is in the feedback loop

Case 2, we select M and N so that the tracking error $y(t) - r(t)$ is used as an input to the compensator.

Recall,

$$\dot{\hat{\mathbf{x}}}(t) = \mathbf{A}\hat{\mathbf{x}}(t) + \mathbf{B}\tilde{u}(t) + \mathbf{L}\tilde{y}(t) + \mathbf{M}r(t)$$

$$u(t) = \tilde{u}(t) + Nr(t) = -\mathbf{K}\hat{\mathbf{x}}(t) + Nr(t),$$

suppose that we select

$$N = 0 \text{ and } \mathbf{M} = -\mathbf{L}$$

Then, the compensator is given by

$$\dot{\hat{\mathbf{x}}}(t) = \mathbf{A}\hat{\mathbf{x}}(t) + \mathbf{B}u(t) + \mathbf{L}\tilde{y}(t) - \mathbf{L}r(t)$$

$$u(t) = -\mathbf{K}\hat{\mathbf{x}}(t),$$

leads to

$$\dot{e} = \dot{x} - \dot{\hat{x}} = (A - LC)e + Lr$$

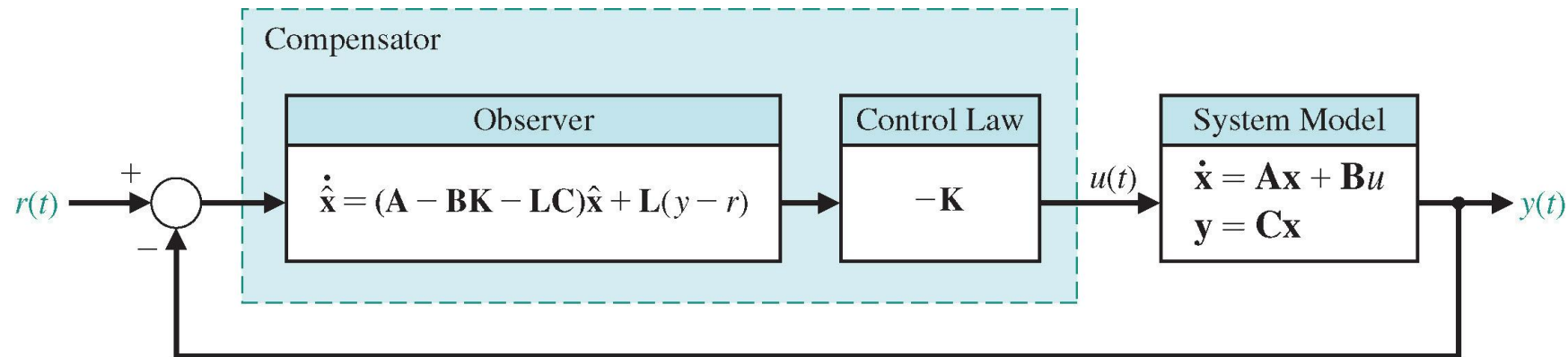
$$\dot{x} = (A - BK)x + BKe$$

Reference Input



Case 2, we select M and N so that the tracking error $y(t) - r(t)$ is used as an input to the compensator.
rewrite the compensator as

$$\begin{aligned}\dot{\hat{\mathbf{x}}}(t) &= (\mathbf{A} - \mathbf{BK} - \mathbf{LC})\hat{\mathbf{x}}(t) + \mathbf{L}(y(t) - r(t)) \\ u(t) &= -\mathbf{K}\hat{\mathbf{x}}(t).\end{aligned}$$



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the compensator is in the forward path.

Depending on the choice of N and M , other implementations are possible, for instance, *the internal model design*.

*Now, we consider the problem of designing a compensator that provides **asymptotic tracking of a reference input with zero steady-state error**.*

└ include steps, ramps, and other persistent signals, such as sinusoids

└ achieved by type-one system, type-two system, and ??

*This idea is formalized here by **introducing an internal model of the reference input in the compensator***

Consider

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t), \quad y(t) = \mathbf{C}\mathbf{x}(t).$$

We consider a reference input to be generated by a linear system of the form

$$\dot{\mathbf{x}}_r(t) = \mathbf{A}_r\mathbf{x}_r(t), \quad r(t) = \mathbf{d}_r\mathbf{x}_r(t),$$

with unknown initial conditions.

For instance, for a step reference input

$$\dot{x}_r(t) = 0, \quad r(t) = x_r(t),$$

Then, the tracking error $e(t)$ is defined as

$$e(t) = y(t) - r(t).$$

Taking the time derivative yields

$$\dot{e}(t) = \dot{y}(t) = \mathbf{C}\dot{\mathbf{x}}(t).$$

If we define the two intermediate variables

$$\mathbf{z}(t) = \dot{\mathbf{x}}(t) \quad \text{and} \quad w(t) = \dot{u}(t),$$

we have

$$\begin{pmatrix} \dot{e}(t) \\ \dot{\mathbf{z}}(t) \end{pmatrix} = \begin{bmatrix} 0 & \mathbf{C} \\ 0 & \mathbf{A} \end{bmatrix} \begin{pmatrix} e(t) \\ \mathbf{z}(t) \end{pmatrix} + \begin{bmatrix} 0 \\ \mathbf{B} \end{bmatrix} w(t).$$

If the system is controllable, we can find a feedback of the form

$$w(t) = -K_1 e(t) - \mathbf{K}_2 \mathbf{z}(t)$$

such that the system is stable.

This implies we will have achieved the objective of asymptotic tracking with zero steady state error.

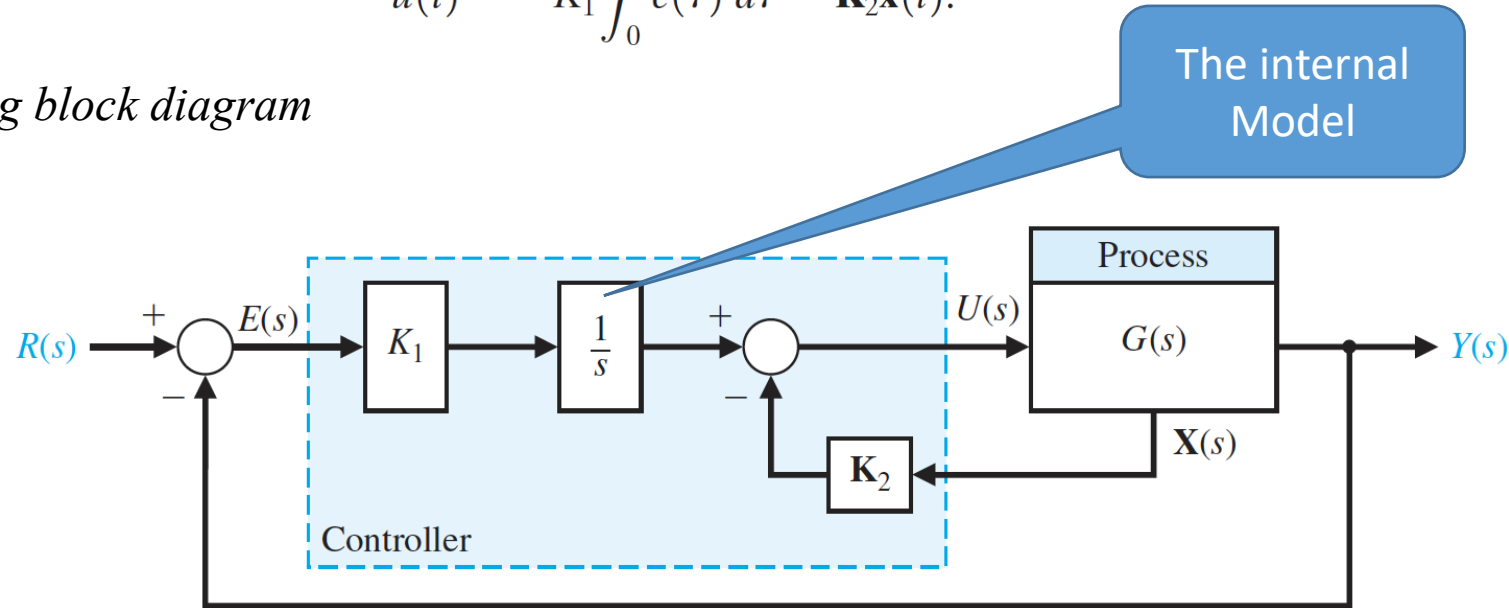
Internal Model Design



The control input, found by

$$u(t) = -K_1 \int_0^t e(\tau) d\tau - \mathbf{K}_2 \mathbf{x}(t).$$

The corresponding block diagram



The internal model principle states that if $G(s)G_c(s)$ contains $R(s)$, then $y(t)$ will track $r(t)$ asymptotically.

achieved by type-one system, type-two system, and controller contains $\frac{\omega}{s^2 + \omega^2}$



Optimal Control (LQR)

***Definition:** The design of a systems that are adjusted to provide a minimum performance index such as*

$$J = \int_0^{\infty} g(\mathbf{x}, \mathbf{u}, t) dt,$$

are called optimal control systems

Consider the LTI SISO system

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t).$$

select a feedback controller as

$$u(t) = -\mathbf{K}\mathbf{x}(t),$$

yields

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) - \mathbf{B}\mathbf{K}\mathbf{x}(t) = \mathbf{H}\mathbf{x}(t),$$

where $\mathbf{H}$ is the $n \times n$ matrix.

Consider an error-squared performance index

$$J = \int_0^{\infty} \mathbf{x}^T(t) \mathbf{x}(t) dt.$$

where

$$\begin{aligned} \mathbf{x}^T(t) \mathbf{x}(t) &= (x_1(t), x_2(t), x_3(t), \dots, x_n(t)) \begin{pmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{pmatrix} \\ &= x_1^2(t) + x_2^2(t) + x_3^2(t) + \dots + x_n^2(t), \end{aligned}$$

To obtain the minimum value of J , we *postulate the existence* of an exact differential so that

$$\frac{d}{dt}(\mathbf{x}^T(t) \mathbf{P} \mathbf{x}(t)) = -\mathbf{x}^T(t) \mathbf{x}(t),$$

where P is to be determined. A symmetric P matrix will be used to simplify the algebra without any loss of generality.

Completing the differentiation indicated on the left-hand side

$$\begin{aligned}\frac{d}{dt}(\mathbf{x}^T(t)\mathbf{P}\mathbf{x}(t)) &= \dot{\mathbf{x}}^T(t)\mathbf{P}\mathbf{x}(t) + \mathbf{x}^T(t)\mathbf{P}\dot{\mathbf{x}}(t). \\ &= \mathbf{x}^T(t)(\mathbf{H}^T\mathbf{P} + \mathbf{P}\mathbf{H})\mathbf{x}(t).\end{aligned}$$

If we let

$$\mathbf{H}^T\mathbf{P} + \mathbf{P}\mathbf{H} = -\mathbf{I},$$

then

$$\frac{d}{dt}(\mathbf{x}^T(t)\mathbf{P}\mathbf{x}(t)) = -\mathbf{x}^T(t)\mathbf{x}(t),$$

If $\mathbf{H}$ is Hurwitz, the existence of an symmetric and positive definite matrix $\mathbf{P}$ is guaranteed, this equation is aka the **Lyapunov equation**

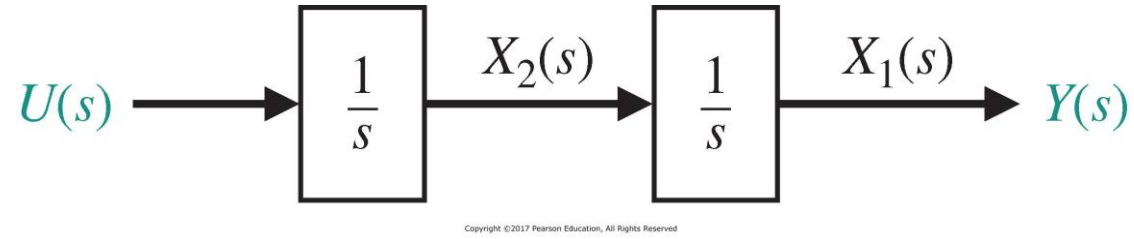
which indicates to

$$J = \int_0^\infty -\frac{d}{dt}(\mathbf{x}^T(t)\mathbf{P}\mathbf{x}(t)) dt = -\mathbf{x}^T(t)\mathbf{P}\mathbf{x}(t) \Big|_0^\infty = \mathbf{x}^T(0)\mathbf{P}\mathbf{x}(0).$$

The design steps are then as follows

1. Determine the matrix $\mathbf{P}$ that satisfies above Lyapunov equation, where $\mathbf{H}$ is known.
2. Minimize J by determining the minimum of $\mathbf{x}^T(0)\mathbf{P}\mathbf{x}(0)$ by adjusting one or more unspecified system parameters.

Example:



The vector differential equation of this system is

$$\frac{d}{dt} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t).$$

We choose a feedback control system so that

$$u(t) = -k_1 x_1(t) - k_2 x_2(t),$$

Then the system becomes

$$\begin{aligned} \dot{x}_1(t) &= x_2(t), \\ \dot{x}_2(t) &= -k_1 x_1(t) - k_2 x_2(t). \end{aligned}$$

Example:

In matrix form, we have

$$\dot{\mathbf{x}}(t) = \mathbf{H}\mathbf{x}(t) = \begin{bmatrix} 0 & 1 \\ -k_1 & -k_2 \end{bmatrix} \mathbf{x}(t).$$

Let $k_1 = 1$ and determine a suitable value for k_2 so that the performance index is minimized.

From the Lyapunov equation, it follows that

$$\begin{bmatrix} 0 & -1 \\ 1 & -k_2 \end{bmatrix} \begin{bmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{bmatrix} + \begin{bmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{bmatrix} \begin{bmatrix} 0 & 1 \\ -1 & -k_2 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Completing the matrix multiplication and addition yields

$$\begin{aligned} -p_{12} - p_{12} &= -1, \\ p_{11} - k_2 p_{12} - p_{22} &= 0, \\ p_{12} - k_2 p_{22} + p_{12} - k_2 p_{22} &= -1. \end{aligned} \quad \Rightarrow \quad p_{12} = \frac{1}{2}, \quad p_{22} = \frac{1}{k_2}, \quad p_{11} = \frac{k_2^2 + 2}{2k_2}.$$

Example:

Consider the integral performance index is then

$$J = \mathbf{x}^T(0) \mathbf{P} \mathbf{x}(0),$$

where

$$\mathbf{x}^T(0) = (1, 1).$$

Therefore J becomes

$$J = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} p_{11} & p_{12} \\ p_{12} & p_{22} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = p_{11} + 2p_{12} + p_{22}.$$

Substituting the values of the elements of P , we have

$$J = \frac{k_2^2 + 2}{2k_2} + 1 + \frac{1}{k_2} = \frac{k_2^2 + 2k_2 + 4}{2k_2}.$$

To minimize as a function of k_2 ,

$$\frac{dJ}{dk_2} = \frac{2k_2(2k_2 + 2) - 2(k_2^2 + 2k_2 + 4)}{(2k_2)^2} = 0.$$

Example:

Therefore

$$k_2 = 2$$

when J is a minimum. The minimum value of J is

$$J_{\min} = 3.$$

The system matrix H , obtained for the compensated system, is then

$$\mathbf{H} = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}.$$

*The **characteristic equation** of the compensated system is therefore*

$$\det[\lambda \mathbf{I} - \mathbf{H}] = \det \begin{bmatrix} \lambda & -1 \\ 1 & \lambda + 2 \end{bmatrix} = \lambda^2 + 2\lambda + 1.$$

therefore the damping ratio of the compensated system is $\xi = 1$

we recognize that this system is optimal only for the specific set of initial conditions that were assumed.

Example continue:

let us consider again

$$\dot{\mathbf{x}}(t) = \mathbf{H}\mathbf{x}(t) = \begin{bmatrix} 0 & 1 \\ -k_1 & -k_2 \end{bmatrix} \mathbf{x}(t).$$

with $k_1 = k_2 = k$. Then system becomes

$$\dot{\mathbf{x}}(t) = \mathbf{H}\mathbf{x}(t) = \begin{bmatrix} 0 & 1 \\ -k & -k \end{bmatrix} \mathbf{x}(t).$$

To determine the P matrix, we use the Lyapunov equation, yielding

$$p_{12} = \frac{1}{2k}, \quad p_{22} = \frac{k+1}{2k^2}, \quad \text{and} \quad p_{11} = \frac{1+2k}{2k}.$$

Let us consider the case

$$\mathbf{x}^T(0) = (1 \quad 0)$$

Example continue:

Then the performance index becomes

$$J = \int_0^{\infty} \mathbf{x}^T(t) \mathbf{x}(t) dt = \mathbf{x}^T(0) \mathbf{P} \mathbf{x}(0) = p_{11} = \frac{1 + 2k}{2k} = 1 + \frac{1}{2k}.$$

Then the minimum value of J is obtained when k approaches infinity, which is 1.

Now, we recognize that, in providing a very large gain k , we can cause the feedback signal

$$u(t) = -k(x_1(t) + x_2(t))$$

to be very large, which is unrealistic, cause in many cases, we have physical limits on the control magnitude.

We can limit the control effort by including it within the expression for the performance index

$$J = \int_0^{\infty} (\mathbf{x}^T(t) \mathbf{I} \mathbf{x}(t) + \lambda \mathbf{u}^T(t) \mathbf{u}(t)) dt,$$

The weighting factor λ will be chosen so that the relative importance of the state variable performance is contrasted with the importance of the control energy.

Example continue:

Now, let us consider again when λ is other than zero and account for the expenditure of control signal energy.

we still use a state variable feedback

$$u(t) = -\mathbf{K}\mathbf{x}(t) = \begin{bmatrix} -k & -k \end{bmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}.$$

The performance function becomes

$$J = \int_0^\infty \mathbf{x}^T(t)(\mathbf{I} + \lambda\mathbf{K}^T\mathbf{K})\mathbf{x}(t) dt = \int_0^\infty \mathbf{x}^T(t)\mathbf{Q}\mathbf{x}(t) dt,$$

$$\mathbf{Q} = \mathbf{I} + \lambda\mathbf{K}^T\mathbf{K} = \begin{bmatrix} 1 + \lambda k^2 & \lambda k^2 \\ \lambda k^2 & 1 + \lambda k^2 \end{bmatrix}.$$

let $\mathbf{x}^T(0) = (1, 0)$ yielding

$$J = p_{11} = (1 + \lambda k^2) \left(1 + \frac{1}{2k} \right) - \lambda k^2. \quad \rightarrow \quad \frac{dJ}{dk} = \frac{1}{2} \left(\lambda - \frac{1}{k^2} \right) = 0.$$

Therefore, the minimum of the performance index occurs when

$$k = k_{\min} = 1/\sqrt{\lambda},$$

Linear Quadratic Regulator (LQR)

Previous design procedure can be carried out for a more general LTI SISO systems

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}u(t)$$

with feedback

$$u(t) = -\mathbf{K}\mathbf{x}(t) = -[k_1 \ k_2 \ \dots \ k_n]\mathbf{x}(t).$$

We can consider the performance index

$$J = \int_0^\infty (\mathbf{x}^T(t)\mathbf{Q}\mathbf{x}(t) + Ru^2(t)) dt,$$

where $R>0$ is a scalar weighting factor. This index is minimized when

$$\mathbf{K} = R^{-1}\mathbf{B}^T\mathbf{P}.$$

The $n \times n$ matrix P is determined from the solution of the equation

$$\mathbf{A}^T\mathbf{P} + \mathbf{P}\mathbf{A} - \mathbf{P}\mathbf{B}R^{-1}\mathbf{B}^T\mathbf{P} + \mathbf{Q} = \mathbf{0}.$$

*which is often called the **algebraic Riccati equation**.*



Kalman Filter (Optional)

Regulator problem

Controllability

Full-state feedback control law

Pole placement

Separation principle

Observability

Observer

Estimation error



Output Regulation

Robust control

Kalman state-space decomposition

Kalman filter

Reduced-order observer design

Adaptive control

Optimal control

MPC

Command following

Internal model design

Linear quadratic regulator